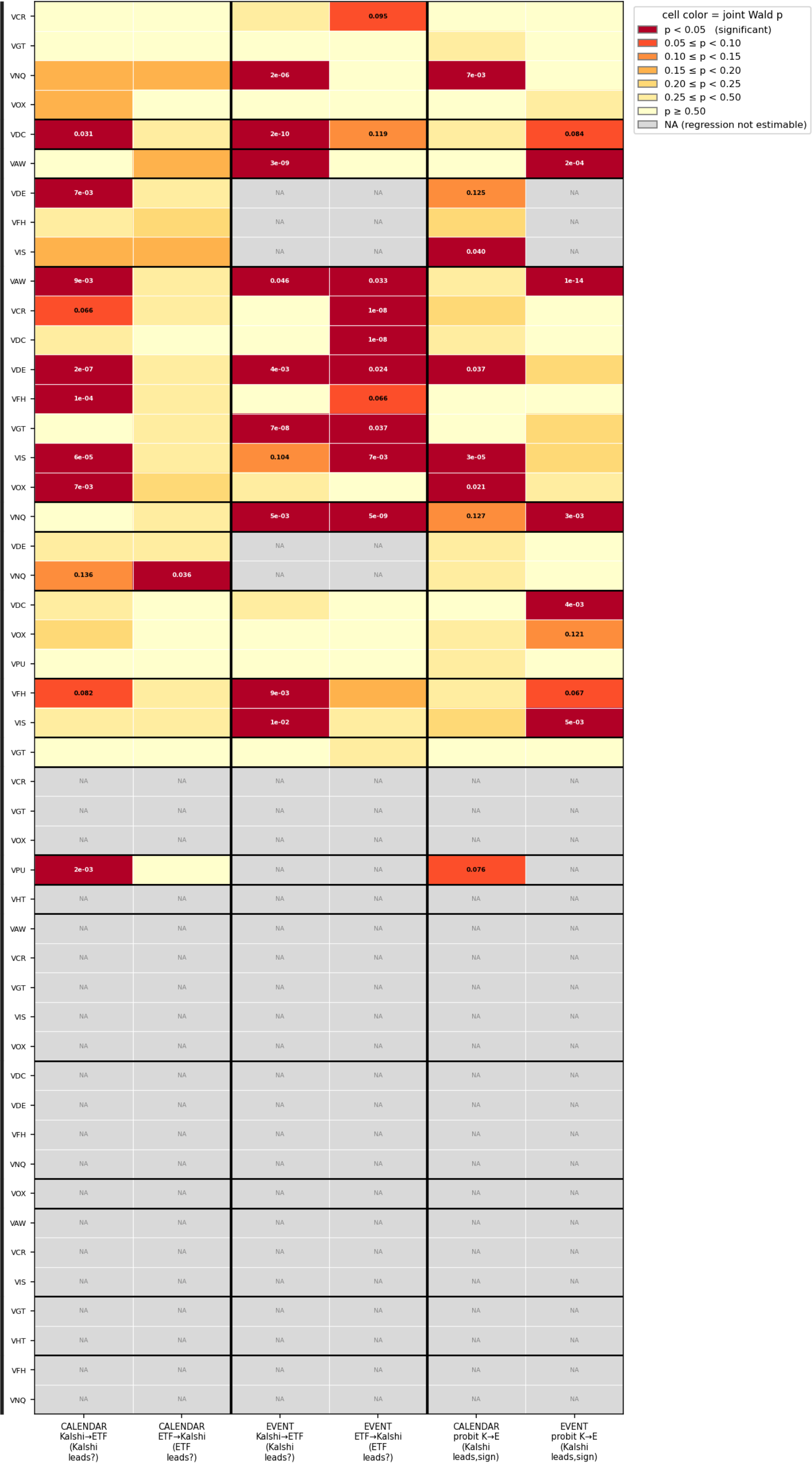


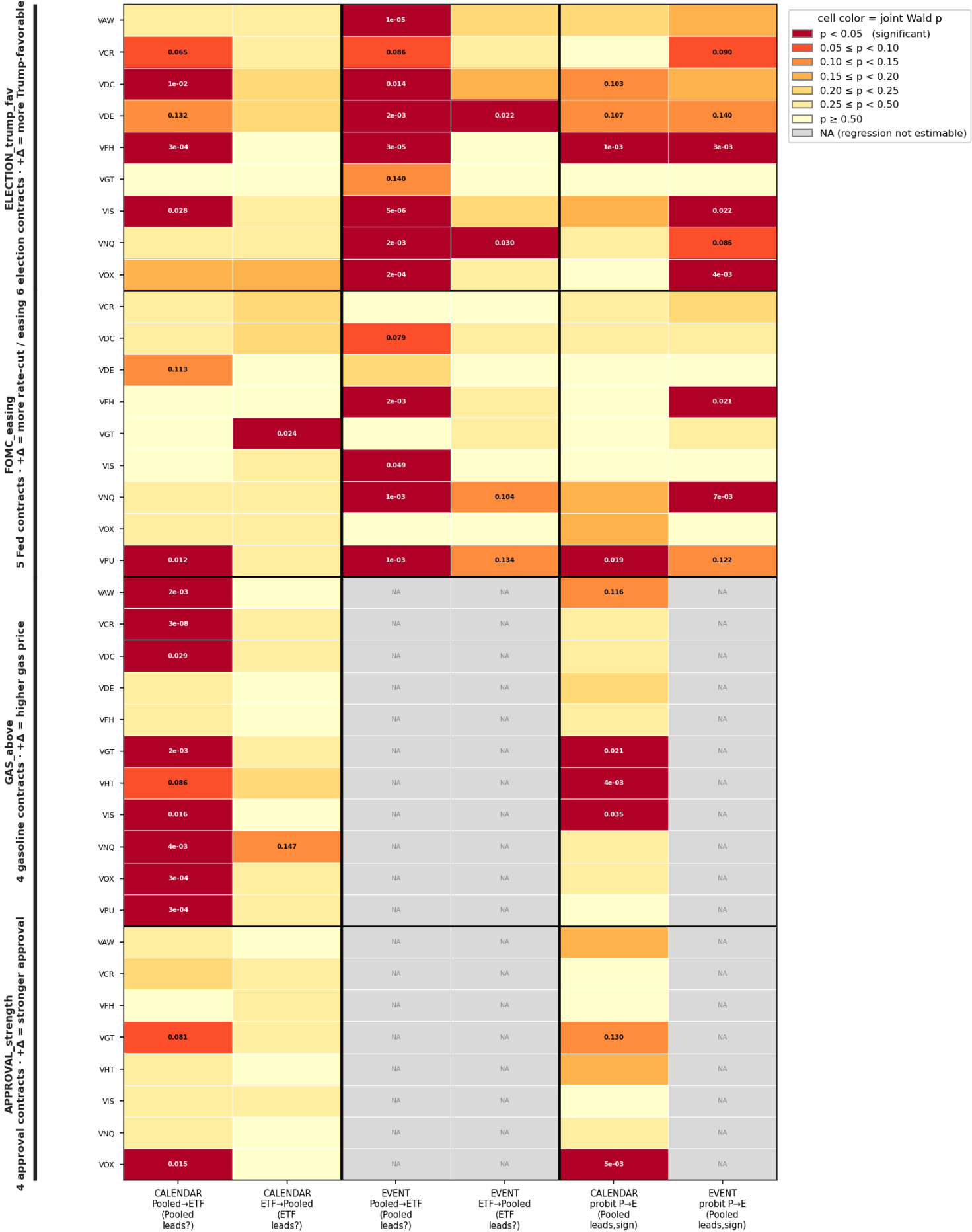
CORE FIGURE 1 — single-pair Granger joint-Wald p heatmap

GRANGER joint-Wald p — all 48 single pairs × 6 regressions, grouped by contract
each cell's number = joint Wald p (H0: NO Granger causality). p<0.05 (red) = reject H0 = that direction's lead/lag IS significant.
rows are grouped by contract (left labels), never sorted by p. 21 all-grey rows = too sparse to estimate any regression.



CORE FIGURE 2 — merged pooled signals Granger heatmap

GRANGER joint-Wald p — merged pooled signals: each group × its ETFs × 6 regressions
Pooled = pooled sign-aligned member contracts (each left block = one pooled signal, labelled). cell = joint Wald p (H0: NO Granger causality); p<0.05 (red) = that direction's lead IS significant.
CALENDAR Pooled→ETF lit while ETF→Pooled pale ⇒ the pooled signal Granger-leads the ETF.



COVERAGE – why 27 of 48 pairs carry results

COVERAGE – 27 of 48 pairs carry Granger results; 21 are dropped (shown as NA rows in core figure 1).

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WHY 21 PAIRS ARE DROPPED (all fail at the SAME step – not a model failure, a data-density failure):

The joint-Wald test needs enough **price-changing** observations to fit the smallest admissible model (K cause-lags + BIC own-lags + day fixed effects) and still have residual degrees of freedom for a valid Newey-West (hac-panel) covariance. ONE gate, applied identically to every pair, every time-axis, and BOTH specifications, BEFORE any regression is attempted:

$$n_{\text{active}} = \# \text{ observations where the cause actually moved } \geq 2K + 5 = 11 \quad (\text{at } K = 3)$$

The threshold is derived from the model's parameter count (K cause-lags plus own-lags plus intercept), not chosen by hand. linear and probit share the same sample (the same calendar grid / event sequence) and the same gate, so a pair is never estimable under one specification but not the other. probit additionally needs both up- and down-moves to appear in the dependent variable, which is an intrinsic requirement of the model rather than a discretionary cutoff.

Every dropped pair fails this gate: the Kalshi contract barely traded during its overlap window, so after building bars almost no bar carries a non-zero price change. There is simply not enough independent information to identify a lag polynomial, let alone test it. Nothing is estimated and the row is left NA rather than reporting an under-identified / rank-deficient p-value.

Of the 21 dropped, 0 had ZERO price-changing bars in the window (no trading / no quote updates at all); the rest had only a handful, well under 11.

DROPPED PAIRS (price-changing observations, measured on the same grid/sequence used for estimation):

contract	etf	cal	evt	reason
Approval >43% by Oct31	VOX	2	2	too few price-changing observations (< 11)
Gas >\$3.30 by Nov30	VDC	2	2	too few price-changing observations (< 11)
Gas >\$3.30 by Nov30	VDE	2	2	too few price-changing observations (< 11)
Gas >\$3.30 by Nov30	VFH	2	2	too few price-changing observations (< 11)
Gas >\$3.30 by Nov30	VNQ	2	2	too few price-changing observations (< 11)
Approval >43% by Sep30	VAW	3	3	too few price-changing observations (< 11)
Approval >43% by Sep30	VCR	3	3	too few price-changing observations (< 11)
Approval >43% by Sep30	VIS	3	3	too few price-changing observations (< 11)
Gas >\$3.20 by Oct31	VHT	3	3	too few price-changing observations (< 11)
Approval >41% by Nov30	VGT	3	3	too few price-changing observations (< 11)
Approval >41% by Nov30	VHT	3	3	too few price-changing observations (< 11)
Gas >\$3.15 by Sep30	VAW	5	5	too few price-changing observations (< 11)
Gas >\$3.15 by Sep30	VCR	5	5	too few price-changing observations (< 11)
Gas >\$3.15 by Sep30	VGT	5	5	too few price-changing observations (< 11)
Gas >\$3.15 by Sep30	VIS	5	5	too few price-changing observations (< 11)
Gas >\$3.15 by Sep30	VOX	5	5	too few price-changing observations (< 11)
Approval <37% by Nov30	VFH	8	8	too few price-changing observations (< 11)
Approval <37% by Nov30	VNQ	8	8	too few price-changing observations (< 11)
Fed cut by Sep-24	VCR	8	8	too few price-changing observations (< 11)
Fed cut by Sep-24	VGT	8	8	too few price-changing observations (< 11)
Fed cut by Sep-24	VOX	8	8	too few price-changing observations (< 11)

TAKEAWAY: the 27 estimable pairs are exactly those liquid enough to support valid inference; the drop is a mechanical minimum-sample rule, applied identically to every pair and both specifications, not a selection on results.

GRANGER JOINT CAUSALITY — Kalshi prediction markets vs Vanguard sector ETFs

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This is the HEADLINE statistic: one joint Wald test per direction giving ONE p-value, replacing the per-coefficient counting used in the main lead-lag report. It is the high-frequency reinstatement of this project's daily Var.py results.test_causality (a VAR Granger Wald), and the object the advisor asked for. Full method: GRANGER_TEST.md.

DEFINITIONS

x_t = Kalshi yes-probability change (Δ prob) over bar t ; y_t = ETF log return over bar t.
Only PAST lags of the 'cause' are used ($j \geq 1$); the contemporaneous ($j=0$) and future ($j < 0$) terms are NOT part of the test. Each regression controls for the dependent variable's OWN past (ADL self-lags, order chosen by BIC).

THE TEST (per direction)

Kalshi→ETF: $y_t = \alpha + \sum_{i=1..p} \phi_i \cdot y_{t-i} + \sum_{j=1..K} b_j \cdot x_{t-j} + \text{day-FE}$; $H_0: b_1=...=b_K=0$
ETF→Kalshi: $x_t = \alpha + \sum_{i=1..q} \psi_i \cdot x_{t-i} + \sum_{j=1..K} d_j \cdot y_{t-j} + \text{day-FE}$; $H_0: d_1=...=d_K=0$
Reject H_0 ($p < 0.05$) $\Rightarrow$ that variable Granger-leads the other. Comparing the two p's states who leads.

STANDARD ERRORS & VALIDITY

Joint Wald uses panel Newey-West (hac-panel, bandwidth=K, within each trading-day block): full-rank (valid even for single-day event contracts, where day-clustering would be rank-deficient) and the microstructure-standard covariance for high-frequency intraday data. Δ prob and log-returns are differenced $\Rightarrow$ stationary $\Rightarrow$ plain Granger is valid, no cointegration/VECM needed.

SAMPLE GATE (identical for both specifications)

A regression is attempted only if n_active — the number of observations where the CAUSE actually moved — is at least 2K+5 (=11 at K=3), a bound derived from the model's parameter count. linear and probit use the SAME sample (same calendar grid / event sequence) and the SAME gate, so no pair is estimable under one specification but not the other. See the COVERAGE section.

MULTIPLE TESTING

Each (time-axis $\times$ specification $\times$ direction) is one test family; within it the joint-Wald p-values are Benjamini-Hochberg FDR-corrected across all pairs (column p_fdr). Both raw p and FDR p are reported; conclusions rest on the FDR-corrected counts and, above all, on the asymmetry between directions.

CONVENTIONS: ET | 09:30-16:00 | ETF log return | median causal bars | day-grouped lags (no overnight)
| adaptive bar & K by contract activity | ADL self-lags by BIC | significance $p < 0.05$.

CONCLUSIONS

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HEADLINE — the direction is asymmetric. Whenever Granger causality is significant it runs Kalshi $\rightarrow$ ETF; the reverse (ETF $\rightarrow$ Kalshi) is essentially absent. This asymmetry — not the raw count of significant pairs — is the robust finding, and it answers the project's question: the evidence supports prediction markets LEADING sector ETFs, and does not support the reverse.

1. Direction is one-way (Kalshi $\rightarrow$ ETF)

- Single pairs, calendar: 8 Kalshi-leads vs 1 ETF-leads (0 bidir, 18 neither, of 27).
- Single pairs, event: 5 Kalshi-leads vs 3 ETF-leads (4 bidir, 9 neither, of 21).
- Pooled signals, calendar: 13 Pooled-leads vs 1 ETF-leads (0 bidir, 23 neither, of 37).
- Pooled signals, event: 9 Pooled-leads vs 0 ETF-leads (2 bidir, 7 neither, of 18).

The reverse direction is significant in ~0 cases at every level — the lead is one-directional.

2. Weak and concentrated at the single-contract level

- 18 of 27 single pairs show no Granger causality either way (calendar). The 8 that do are concentrated in a handful of contracts — mainly 'Harris 276 EV' across 5 ETFs, plus 3 other contracts with one ETF each:

Harris 276 EV×VDE, Harris 276 EV×VIS, Harris 276 EV×VFH, Gas >\$3.15 by Oct31×VPU, Trump 316 EV×VDE, Harris 276 EV×VOX, Harris 276 EV×VAW, Trump 306 EV×VDC.

- So the lead is NOT a broad per-contract phenomenon; it is concentrated in the few LIQUID contracts. Illiquid contracts have no usable price-discovery process (consistent with the advisor's point).

3. Pooling restores power

- Aggregating same-family contracts into sign-aligned pooled signals (more degrees of freedom) lights up Pooled→ETF in 13/37 (calendar) and 9/18 (event), while the reverse stays ~0. The lead is real but needs aggregation to detect above single-contract noise.
- Per pooled signal, Pooled→ETF significance (raw $p < 0.05$, then after BH-FDR):

APPROVAL_strength	calendar: 1/8 raw, 0 after FDR		event: not estimable
ELECTION_trump_fav	calendar: 3/9 raw, 2 after FDR		event: 7/9 raw, 7 after FDR
FOMC_easing	calendar: 1/9 raw, 1 after FDR		event: 4/9 raw, 3 after FDR
GAS_above	calendar: 8/11 raw, 6 after FDR		event: not estimable

Only the pooled signals with enough ACTIVE event-time observations are estimable on the event axis; the others show 'not estimable' there rather than a null result.

4. Caveats (state them)

- Time-axis: the clean one-way result is on the CALENDAR (clock-time) axis. EVENT-time is muddier (3 single-pair ETF-leads, 4 bidirectional), most likely an artifact of sampling only at the sparse Kalshi trade instants (near-contemporaneous comovement misread as ETF-leads). The headline rests on calendar; event-time is robustness / a warning, not the basis for the claim.
- Direction-only (probit) is weak: 5/27 (calendar) and 5/23 (event) significant. Predictability is in the MAGNITUDE of continuous returns more than in the up/down sign.

5. Multiple testing — the asymmetry SURVIVES a Benjamini-Hochberg FDR correction

Every direction×axis×spec is one test family; each family is BH-FDR corrected across all pairs, so "8 of 27 significant at 5%" cannot be read as evidence on its own (~1.4 would be expected by chance). Counts below are raw $p < 0.05 \rightarrow$ FDR $p < 0.05$:

- Single pairs, calendar: Kalshi→ETF 8→7 of 27 | ETF→Kalshi 1→0 of 27
- Single pairs, event: Kalshi→ETF 9→8 of 21 | ETF→Kalshi 7→4 of 21
- Pooled signals, calendar: Pooled→ETF 13→9 of 37 | ETF→Pooled 1→0 of 37
- Pooled signals, event: Pooled→ETF 11→10 of 18 | ETF→Pooled 2→0 of 18

The Kalshi→ETF direction largely survives FDR, while the reverse direction collapses to ~0 once corrected. So the one-way lead is not an artifact of running many tests: correcting for multiplicity REMOVES the reverse-direction hits and KEEPS the forward ones.

BOTTOM LINE — High-frequency Granger tests show prediction markets one-directionally lead sector ETFs on the clock-time axis (reverse essentially absent); the lead is concentrated in a few high-liquidity contracts and is most robust after pooling same-family contracts into pooled signals; direction-only (probit) and event-time evidence is weaker. 'Lead' here means Granger precedence (predictive), not cause.

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SINGLE-PAIR SUMMARY — grouped by contract (not sorted by p).
joint-Wald p per direction; * <.10 ** <.05 *** <.01; '+' = still significant after BH-FDR.

contract	etf	cal K→E	cal E→K	evt K→E	evt E→K	verdict(cal/evt)
Trump 281 EV	VCR	0.9	0.87	0.3	0.095*	-/-
	VGT	0.65	0.64	0.78	0.59	-/-
	VNQ	0.16	0.19	2e-06***†	0.56	-/K→E
	VOX	0.16	0.77	0.6	0.69	-/-
Trump 306 EV	VDC	0.031**	0.34	2.1e-10***†	0.12	K→E/K→E
Trump 312 EV	VAW	0.97	0.19	3.2e-09***†	0.66	-/K→E
Trump 316 EV	VDE	0.0067***†	0.49	-	-	K→E/-
	VFH	0.37	0.23	-	-	-/-
	VIS	0.17	0.19	-	-	-/-
Harris 276 EV	VAW	0.0089***†	0.34	0.046**	0.033**	K→E/bi
	VCR	0.066*	0.42	0.59	1.1e-08***†	-/E→K
	VDC	0.31	0.63	0.84	1.5e-08***†	-/E→K
	VDE	2.5e-07***†	0.41	0.0039***†	0.024**	K→E/bi
	VFH	0.00013***†	0.32	0.65	0.066*	K→E/-
	VGT	0.53	0.35	6.7e-08***†	0.037**	-/bi
	VIS	6e-05***†	0.31	0.1	0.0068***†	K→E/E→K
	VOX	0.0075***†	0.23	0.38	0.86	K→E/-
Harris 287 EV	VNQ	0.65	0.33	0.0053***†	4.5e-09***†	-/bi
Fed Nov-24 hold 0bp	VDE	0.4	0.38	-	-	-/-
	VNQ	0.14	0.036**	-	-	E→K/-
Fed Sep-24 cut 25bp	VDC	0.48	0.75	0.37	0.74	-/-
	VOX	0.2	0.71	0.89	0.64	-/-
	VPU	0.88	0.59	0.84	0.54	-/-
Fed Dec-24 cut 25bp	VFH	0.082*	0.49	0.0092***†	0.15	-/K→E
	VIS	0.46	0.25	0.0099***†	0.43	-/K→E
Fed Dec-24 hold 0bp	VGT	0.59	0.88	0.95	0.34	-/-
Gas >\$3.15 by Oct31	VPU	0.0017***†	0.58	-	-	K→E/-

Fed Dec-24 cut 25bp x VIS [ticker: KXFEDDECISION-24DEC-C25]
Contract : "Will the Federal Reserve Cut rates by 25bps at their December 2024 meeting?"
Window : 2024-10-31 to 2024-12-18 primary bar : 5min Kalshi trades : 2308

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 => that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:
 $y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j} + \sum(d=1..32) \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=1 (BIC) ; cause block K=10 ; day-FE over 33 days ; hac-panel SE (bw=10).
H0: $b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 9.82, p = 0.457 \mid \text{BH-FDR } p = 0.649$
Direction B — ETF → Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum(j=1..10) d_j \cdot y_{t-j} + \sum(d=1..32) \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=1 (BIC) ; cause block K=10 ; day-FE over 33 days ; hac-panel SE (bw=10).
H0: $d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 12.55, p = 0.25 \mid \text{BH-FDR } p = 0.706$
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j})$
controls: ADL self-lags p=1 (BIC) ; cause block K=10 ; sample spans 33 days (probit: no day-FE) ; hac-panel SE (bw=10).
H0: $b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 13.15, p = 0.215 \mid \text{BH-FDR } p = 0.586$
=> calendar verdict: neither direction significant.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi → ETF:
 $y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..10) b_j \cdot x_{t-j} + \sum(d=1..27) \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=0 (BIC) ; cause block K=10 ; day-FE over 28 days ; hac-panel SE (bw=10).
H0: $b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 23.24, p = 0.00989 \text{ ***} \mid \text{BH-FDR } p = 0.0259 \text{ **}$
Direction B — ETF → Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \psi_6 \cdot x_{t-6} + \sum(j=1..10) d_j \cdot y_{t-j} + \sum(d=1..27) \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=6 (BIC) ; cause block K=10 ; day-FE over 28 days ; hac-panel SE (bw=10).
H0: $d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 10.14, p = 0.428 \mid \text{BH-FDR } p = 0.692$
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..10) b_j \cdot x_{t-j})$
controls: ADL self-lags p=0 (BIC) ; cause block K=10 ; sample spans 28 days (probit: no day-FE) ; hac-panel SE (bw=10).
H0: $b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 25.25, p = 0.00489 \text{ ***} \mid \text{BH-FDR } p = 0.0225 \text{ **}$
=> event verdict: Kalshi Granger-leads ETF.

3. OVERALL VERDICT

Calendar says 'neither direction significant' but Event says 'Kalshi Granger-leads ETF' — not robust across time-axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

Fed Dec-24 cut 25bp x VFH [ticker: KXFEDDECISION-24DEC-C25]
Contract : "Will the Federal Reserve Cut rates by 25bps at their December 2024 meeting?"
Window : 2024-10-31 to 2024-12-18 primary bar : 5min Kalshi trades : 2308

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 => that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:
 $y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..10} b_j \cdot x_{t-j} + \sum_{d=1..32} \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=1 (BIC) ; cause block K=10 ; day-FE over 33 days ; hac-panel SE (bw=10).
H0: $b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 16.67, \quad p = 0.0819 * \quad | \quad \text{BH-FDR } p = 0.221$
Direction B — ETF → Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum_{j=1..10} d_j \cdot y_{t-j} + \sum_{d=1..32} \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=1 (BIC) ; cause block K=10 ; day-FE over 33 days ; hac-panel SE (bw=10).
H0: $d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 9.41, \quad p = 0.494 \quad | \quad \text{BH-FDR } p = 0.741$
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..10} b_j \cdot x_{t-j})$
controls: ADL self-lags p=1 (BIC) ; cause block K=10 ; sample spans 33 days (probit: no day-FE) ; hac-panel SE (bw=10).
H0: $b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 11.31, \quad p = 0.334 \quad | \quad \text{BH-FDR } p = 0.586$
=> calendar verdict: neither direction significant.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi → ETF:
 $y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum_{j=1..10} b_j \cdot x_{t-j} + \sum_{d=1..27} \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=0 (BIC) ; cause block K=10 ; day-FE over 28 days ; hac-panel SE (bw=10).
H0: $b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 23.44, \quad p = 0.00924 *** \quad | \quad \text{BH-FDR } p = 0.0259 **$
Direction B — ETF → Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \psi_6 \cdot x_{t-6} + \sum_{j=1..10} d_j \cdot y_{t-j} + \sum_{d=1..27} \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=6 (BIC) ; cause block K=10 ; day-FE over 28 days ; hac-panel SE (bw=10).
H0: $d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 14.47, \quad p = 0.153 \quad | \quad \text{BH-FDR } p = 0.291$
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum_{j=1..10} b_j \cdot x_{t-j})$
controls: ADL self-lags p=0 (BIC) ; cause block K=10 ; sample spans 28 days (probit: no day-FE) ; hac-panel SE (bw=10).
H0: $b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 17.36, \quad p = 0.0667 * \quad | \quad \text{BH-FDR } p = 0.256$
=> event verdict: Kalshi Granger-leads ETF.

3. OVERALL VERDICT

Calendar says 'neither direction significant' but Event says 'Kalshi Granger-leads ETF' — not robust across time-axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

Fed Dec-24 hold 0bp x VGT [ticker: KXFEDDECISION-24DEC-H0]
Contract : "Will the Federal Reserve Hike rates by 0bps at their December 2024 meeting?"
Window : 2024-10-31 to 2024-12-18 primary bar : 10min Kalshi trades : 1375

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 ⇒ that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:
y_t = α + (no y self-lag term: BIC chose p=0) + Σ(j=1..8) b_j·x_{t-j} + Σ(d=1..33) γ_d·Day_d
controls: ADL self-lags p=0 (BIC) ; cause block K=8 ; day-FE over 34 days ; hac-panel SE (bw=8).
H0: b₁ = ... = b₈ = 0 → Wald χ²(8) = 6.47, p = 0.595 | BH-FDR p = 0.727

Direction B — ETF → Kalshi:
x_t = α + ψ₁·x_{t-1} + ψ₂·x_{t-2} + ψ₃·x_{t-3} + ψ₄·x_{t-4} + Σ(j=1..8) d_j·y_{t-j} + Σ(d=1..33) γ_d·Day_d
controls: ADL self-lags p=4 (BIC) ; cause block K=8 ; day-FE over 34 days ; hac-panel SE (bw=8).
H0: d₁ = ... = d₈ = 0 → Wald χ²(8) = 3.77, p = 0.878 | BH-FDR p = 0.878

Direction A (probit, ETF up/down):
Pr(y_t>0) = Φ(α + (no y self-lag term: BIC chose p=0) + Σ(j=1..8) b_j·x_{t-j})
controls: ADL self-lags p=0 (BIC) ; cause block K=8 ; sample spans 34 days (probit: no day-FE) ; hac-panel SE (bw=8).
H0: b₁ = ... = b₈ = 0 → Wald χ²(8) = 6.79, p = 0.559 | BH-FDR p = 0.686
=> calendar verdict: neither direction significant.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi → ETF:
y_t = α + φ₁·y_{t-1} + Σ(j=1..8) b_j·x_{t-j} + Σ(d=1..25) γ_d·Day_d
controls: ADL self-lags p=1 (BIC) ; cause block K=8 ; day-FE over 26 days ; hac-panel SE (bw=8).
H0: b₁ = ... = b₈ = 0 → Wald χ²(8) = 2.76, p = 0.949 | BH-FDR p = 0.949

Direction B — ETF → Kalshi:
x_t = α + ψ₁·x_{t-1} + ψ₂·x_{t-2} + Σ(j=1..8) d_j·y_{t-j} + Σ(d=1..25) γ_d·Day_d
controls: ADL self-lags p=2 (BIC) ; cause block K=8 ; day-FE over 26 days ; hac-panel SE (bw=8).
H0: d₁ = ... = d₈ = 0 → Wald χ²(8) = 9.06, p = 0.337 | BH-FDR p = 0.59

Direction A (probit, ETF up/down):
Pr(y_t>0) = Φ(α + (no y self-lag term: BIC chose p=0) + Σ(j=1..8) b_j·x_{t-j})
controls: ADL self-lags p=0 (BIC) ; cause block K=8 ; sample spans 26 days (probit: no day-FE) ; hac-panel SE (bw=8).
H0: b₁ = ... = b₈ = 0 → Wald χ²(8) = 6.65, p = 0.575 | BH-FDR p = 0.811
=> event verdict: neither direction significant.

3. OVERALL VERDICT

No Granger causality detected on either axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

Fed Sep-24 cut 25bp x VOX [ticker: FEDDECISION-24SEP-C25]
Contract : "Will the Federal Reserve Cut rates by 25bps at their September 2024 meeting?"
Window : 2024-09-04 to 2024-09-18 primary bar : 2min Kalshi trades : 1125

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 ⇒ that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:
 $y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..8) b_j \cdot x_{t-j} + \sum(d=1..10) \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=1 (BIC) ; cause block K=8 ; day-FE over 11 days ; hac-panel SE (bw=8).
H0: $b_1 = \dots = b_8 = 0 \rightarrow \text{Wald } \chi^2(8) = 10.95, p = 0.205 \mid \text{BH-FDR } p = 0.368$
Direction B — ETF → Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \sum(j=1..8) d_j \cdot y_{t-j} + \sum(d=1..10) \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=4 (BIC) ; cause block K=8 ; day-FE over 11 days ; hac-panel SE (bw=8).
H0: $d_1 = \dots = d_8 = 0 \rightarrow \text{Wald } \chi^2(8) = 5.44, p = 0.71 \mid \text{BH-FDR } p = 0.828$
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..8) b_j \cdot x_{t-j})$
controls: ADL self-lags p=1 (BIC) ; cause block K=8 ; sample spans 11 days (probit: no day-FE) ; hac-panel SE (bw=8).
H0: $b_1 = \dots = b_8 = 0 \rightarrow \text{Wald } \chi^2(8) = 8.14, p = 0.42 \mid \text{BH-FDR } p = 0.586$
⇒ calendar verdict: neither direction significant.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi → ETF:
 $y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..8) b_j \cdot x_{t-j} + \sum(d=1..9) \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=0 (BIC) ; cause block K=8 ; day-FE over 10 days ; hac-panel SE (bw=8).
H0: $b_1 = \dots = b_8 = 0 \rightarrow \text{Wald } \chi^2(8) = 3.63, p = 0.889 \mid \text{BH-FDR } p = 0.933$
Direction B — ETF → Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \sum(j=1..8) d_j \cdot y_{t-j} + \sum(d=1..9) \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=2 (BIC) ; cause block K=8 ; day-FE over 10 days ; hac-panel SE (bw=8).
H0: $d_1 = \dots = d_8 = 0 \rightarrow \text{Wald } \chi^2(8) = 6.08, p = 0.638 \mid \text{BH-FDR } p = 0.766$
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..8) b_j \cdot x_{t-j})$
controls: ADL self-lags p=1 (BIC) ; cause block K=8 ; sample spans 10 days (probit: no day-FE) ; hac-panel SE (bw=8).
H0: $b_1 = \dots = b_8 = 0 \rightarrow \text{Wald } \chi^2(8) = 12.73, p = 0.121 \mid \text{BH-FDR } p = 0.349$
⇒ event verdict: neither direction significant.

3. OVERALL VERDICT

No Granger causality detected on either axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

Fed Sep-24 cut 25bp x VPU [ticker: FEDDECISION-24SEP-C25]
Contract : "Will the Federal Reserve Cut rates by 25bps at their September 2024 meeting?"
Window : 2024-09-04 to 2024-09-18 primary bar : 2min Kalshi trades : 1125

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 ⇒ that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:
 $y_t = \alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \phi_3 \cdot y_{t-3} + \phi_4 \cdot y_{t-4} + \phi_5 \cdot y_{t-5} + \phi_6 \cdot y_{t-6} + \sum_{j=1..8} b_j \cdot x_{t-j} + \sum_{d=1..10} \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=6 (BIC) ; cause block K=8 ; day-FE over 11 days ; hac-panel SE (bw=8).
H0: $b_1 = \dots = b_8 = 0 \rightarrow \text{Wald } \chi^2(8) = 3.75, p = 0.879 \mid \text{BH-FDR } p = 0.938$
Direction B — ETF → Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \sum_{j=1..8} d_j \cdot y_{t-j} + \sum_{d=1..10} \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=4 (BIC) ; cause block K=8 ; day-FE over 11 days ; hac-panel SE (bw=8).
H0: $d_1 = \dots = d_8 = 0 \rightarrow \text{Wald } \chi^2(8) = 6.53, p = 0.588 \mid \text{BH-FDR } p = 0.779$
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..8} b_j \cdot x_{t-j})$
controls: ADL self-lags p=1 (BIC) ; cause block K=8 ; sample spans 11 days (probit: no day-FE) ; hac-panel SE (bw=8).
H0: $b_1 = \dots = b_8 = 0 \rightarrow \text{Wald } \chi^2(8) = 10.05, p = 0.262 \mid \text{BH-FDR } p = 0.586$
=> calendar verdict: neither direction significant.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi → ETF:
 $y_t = \alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \sum_{j=1..8} b_j \cdot x_{t-j} + \sum_{d=1..9} \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=2 (BIC) ; cause block K=8 ; day-FE over 10 days ; hac-panel SE (bw=8).
H0: $b_1 = \dots = b_8 = 0 \rightarrow \text{Wald } \chi^2(8) = 4.23, p = 0.836 \mid \text{BH-FDR } p = 0.932$
Direction B — ETF → Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \sum_{j=1..8} d_j \cdot y_{t-j} + \sum_{d=1..9} \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=2 (BIC) ; cause block K=8 ; day-FE over 10 days ; hac-panel SE (bw=8).
H0: $d_1 = \dots = d_8 = 0 \rightarrow \text{Wald } \chi^2(8) = 6.97, p = 0.54 \mid \text{BH-FDR } p = 0.766$
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum_{j=1..8} b_j \cdot x_{t-j})$
controls: ADL self-lags p=0 (BIC) ; cause block K=8 ; sample spans 10 days (probit: no day-FE) ; hac-panel SE (bw=8).
H0: $b_1 = \dots = b_8 = 0 \rightarrow \text{Wald } \chi^2(8) = 6.11, p = 0.635 \mid \text{BH-FDR } p = 0.811$
=> event verdict: neither direction significant.

3. OVERALL VERDICT

No Granger causality detected on either axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

Fed Sep-24 cut 25bp x VDC [ticker: FEDDECISION-24SEP-C25]
Contract : "Will the Federal Reserve Cut rates by 25bps at their September 2024 meeting?"
Window : 2024-09-04 to 2024-09-18 primary bar : 2min Kalshi trades : 1125

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 ⇒ that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:
 $y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..8} b_j \cdot x_{t-j} + \sum_{d=1..10} \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=1 (BIC) ; cause block K=8 ; day-FE over 11 days ; hac-panel SE (bw=8).
H0: $b_1 = \dots = b_8 = 0 \rightarrow \text{Wald } \chi^2(8) = 7.50, p = 0.484 \mid \text{BH-FDR } p = 0.653$
Direction B — ETF → Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \sum_{j=1..8} d_j \cdot y_{t-j} + \sum_{d=1..10} \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=4 (BIC) ; cause block K=8 ; day-FE over 11 days ; hac-panel SE (bw=8).
H0: $d_1 = \dots = d_8 = 0 \rightarrow \text{Wald } \chi^2(8) = 5.07, p = 0.75 \mid \text{BH-FDR } p = 0.828$
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..8} b_j \cdot x_{t-j})$
controls: ADL self-lags p=1 (BIC) ; cause block K=8 ; sample spans 11 days (probit: no day-FE) ; hac-panel SE (bw=8).
H0: $b_1 = \dots = b_8 = 0 \rightarrow \text{Wald } \chi^2(8) = 6.44, p = 0.598 \mid \text{BH-FDR } p = 0.702$
⇒ calendar verdict: neither direction significant.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi → ETF:
 $y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum_{j=1..8} b_j \cdot x_{t-j} + \sum_{d=1..9} \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=0 (BIC) ; cause block K=8 ; day-FE over 10 days ; hac-panel SE (bw=8).
H0: $b_1 = \dots = b_8 = 0 \rightarrow \text{Wald } \chi^2(8) = 8.73, p = 0.366 \mid \text{BH-FDR } p = 0.607$
Direction B — ETF → Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \sum_{j=1..8} d_j \cdot y_{t-j} + \sum_{d=1..9} \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=2 (BIC) ; cause block K=8 ; day-FE over 10 days ; hac-panel SE (bw=8).
H0: $d_1 = \dots = d_8 = 0 \rightarrow \text{Wald } \chi^2(8) = 5.14, p = 0.742 \mid \text{BH-FDR } p = 0.78$
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..8} b_j \cdot x_{t-j})$
controls: ADL self-lags p=1 (BIC) ; cause block K=8 ; sample spans 10 days (probit: no day-FE) ; hac-panel SE (bw=8).
H0: $b_1 = \dots = b_8 = 0 \rightarrow \text{Wald } \chi^2(8) = 22.28, p = 0.00442 \text{ ***} \mid \text{BH-FDR } p = 0.0225 \text{ **}$
⇒ event verdict: neither direction significant.

3. OVERALL VERDICT

No Granger causality detected on either axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

Fed Nov-24 hold 0bp x VNQ [ticker: FEDDECISION-24NOV-H0]
Contract : "Will the Federal Reserve Hike rates by 0bps at their November 2024 meeting?"
Window : 2024-09-18 to 2024-11-07 primary bar : 10min Kalshi trades : 452

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 => that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:
y_t = α + (no y self-lag term: BIC chose p=0) + Σ(j=1..6) b_j·x_{t-j} + Σ(d=1..35) γ_d·Day_d
controls: ADL self-lags p=0 (BIC) ; cause block K=6 ; day-FE over 36 days ; hac-panel SE (bw=6).
H0: b₁ = ... = b₆ = 0 → Wald χ²(6) = 9.75, p = 0.136 | BH-FDR p = 0.324

Direction B — ETF → Kalshi:
x_t = α + (no x self-lag term: BIC chose p=0) + Σ(j=1..6) d_j·y_{t-j} + Σ(d=1..35) γ_d·Day_d
controls: ADL self-lags p=0 (BIC) ; cause block K=6 ; day-FE over 36 days ; hac-panel SE (bw=6).
H0: d₁ = ... = d₆ = 0 → Wald χ²(6) = 13.45, p = 0.0364 ** | BH-FDR p = 0.706

Direction A (probit, ETF up/down):
Pr(y_t>0) = Φ(α + φ₁·y_{t-1} + Σ(j=1..6) b_j·x_{t-j})
controls: ADL self-lags p=1 (BIC) ; cause block K=6 ; sample spans 36 days (probit: no day-FE) ; hac-panel SE (bw=6).
H0: b₁ = ... = b₆ = 0 → Wald χ²(6) = 6.88, p = 0.332 | BH-FDR p = 0.586
=> calendar verdict: ETF Granger-leads Kalshi.

2. EVENT-TIME (event-count lags)

(not estimable at this axis)

3. OVERALL VERDICT

Only one axis significant: ETF Granger-leads Kalshi.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

Fed Nov-24 hold 0bp x VDE [ticker: FEDDECISION-24NOV-H0]
Contract : "Will the Federal Reserve Hike rates by 0bps at their November 2024 meeting?"
Window : 2024-09-18 to 2024-11-07 primary bar : 10min Kalshi trades : 452

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 => that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:
y_t = α + (no y self-lag term: BIC chose p=0) + Σ(j=1..6) b_j·x_{t-j} + Σ(d=1..35) γ_d·Day_d
controls: ADL self-lags p=0 (BIC) ; cause block K=6 ; day-FE over 36 days ; hac-panel SE (bw=6).
H0: b₁ = ... = b₆ = 0 → Wald χ²(6) = 6.25, p = 0.396 | BH-FDR p = 0.594

Direction B — ETF → Kalshi:
x_t = α + (no x self-lag term: BIC chose p=0) + Σ(j=1..6) d_j·y_{t-j} + Σ(d=1..35) γ_d·Day_d
controls: ADL self-lags p=0 (BIC) ; cause block K=6 ; day-FE over 36 days ; hac-panel SE (bw=6).
H0: d₁ = ... = d₆ = 0 → Wald χ²(6) = 6.37, p = 0.383 | BH-FDR p = 0.706

Direction A (probit, ETF up/down):
Pr(y_t>0) = Φ(α + φ₁·y_{t-1} + Σ(j=1..6) b_j·x_{t-j})
controls: ADL self-lags p=1 (BIC) ; cause block K=6 ; sample spans 36 days (probit: no day-FE) ; hac-panel SE (bw=6).
H0: b₁ = ... = b₆ = 0 → Wald χ²(6) = 6.03, p = 0.42 | BH-FDR p = 0.586
=> calendar verdict: neither direction significant.

2. EVENT-TIME (event-count lags)

(not estimable at this axis)

3. OVERALL VERDICT

No Granger causality detected on either axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

=====

Trump 281 EV x VNQ [ticker: KXECDJT281]

Contract : "Will Trump win 281-257 - AZ, GA, NC, PA?"

Window : 2024-11-04 to 2024-11-25 primary bar : 5min Kalshi trades : 308

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).

p<0.05 ⇒ that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:

$y_t = \alpha + \varphi_1 \cdot y_{t-1} + \sum(j=1..5) b_j \cdot x_{t-j} + \sum(d=1..10) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=1 (BIC) ; cause block K=5 ; day-FE over 11 days ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 7.85, \quad p = 0.165 \quad | \quad \text{BH-FDR } p = 0.324$

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \sum(j=1..5) d_j \cdot y_{t-j} + \sum(d=1..10) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=2 (BIC) ; cause block K=5 ; day-FE over 11 days ; hac-panel SE (bw=5).

H0: $d_1 = \dots = d_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 7.40, \quad p = 0.193 \quad | \quad \text{BH-FDR } p = 0.706$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..5) b_j \cdot x_{t-j})$

controls: ADL self-lags p=0 (BIC) ; cause block K=5 ; sample spans 11 days (probit: no day-FE) ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 15.89, \quad p = 0.00718 \text{ ***} \quad | \quad \text{BH-FDR } p = 0.0969 *$

=> calendar verdict: neither direction significant.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi → ETF:

$y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..5) b_j \cdot x_{t-j} + \sum(d=1..3) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=0 (BIC) ; cause block K=5 ; day-FE over 4 days ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 34.36, \quad p = 2.02\text{e-}06 \text{ ***} \quad | \quad \text{BH-FDR } p = 1.06\text{e-}05 \text{ ***}$

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \sum(j=1..5) d_j \cdot y_{t-j} + \sum(d=1..3) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=2 (BIC) ; cause block K=5 ; day-FE over 4 days ; hac-panel SE (bw=5).

H0: $d_1 = \dots = d_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 3.90, \quad p = 0.564 \quad | \quad \text{BH-FDR } p = 0.766$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..5) b_j \cdot x_{t-j})$

controls: ADL self-lags p=0 (BIC) ; cause block K=5 ; sample spans 4 days (probit: no day-FE) ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 2.59, \quad p = 0.763 \quad | \quad \text{BH-FDR } p = 0.895$

=> event verdict: Kalshi Granger-leads ETF.

3. OVERALL VERDICT

Calendar says 'neither direction significant' but Event says 'Kalshi Granger-leads ETF' — not robust across time-axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

=====

Trump 281 EV x VOX [ticker: KXECDJT281]

Contract : "Will Trump win 281-257 - AZ, GA, NC, PA?"

Window : 2024-11-04 to 2024-11-25 primary bar : 5min Kalshi trades : 308

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 ⇒ that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:

$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..5) b_j \cdot x_{t-j} + \sum(d=1..10) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=1 (BIC) ; cause block K=5 ; day-FE over 11 days ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 7.94, \quad p = 0.16 \quad | \quad \text{BH-FDR } p = 0.324$

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \sum(j=1..5) d_j \cdot y_{t-j} + \sum(d=1..10) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=2 (BIC) ; cause block K=5 ; day-FE over 11 days ; hac-panel SE (bw=5).

H0: $d_1 = \dots = d_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 2.56, \quad p = 0.767 \quad | \quad \text{BH-FDR } p = 0.828$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..5) b_j \cdot x_{t-j})$

controls: ADL self-lags p=0 (BIC) ; cause block K=5 ; sample spans 11 days (probit: no day-FE) ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 4.32, \quad p = 0.504 \quad | \quad \text{BH-FDR } p = 0.648$

=> calendar verdict: neither direction significant.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi → ETF:

$y_t = \alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \sum(j=1..5) b_j \cdot x_{t-j} + \sum(d=1..3) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=2 (BIC) ; cause block K=5 ; day-FE over 4 days ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 3.67, \quad p = 0.597 \quad | \quad \text{BH-FDR } p = 0.836$

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \sum(j=1..5) d_j \cdot y_{t-j} + \sum(d=1..3) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=2 (BIC) ; cause block K=5 ; day-FE over 4 days ; hac-panel SE (bw=5).

H0: $d_1 = \dots = d_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 3.05, \quad p = 0.693 \quad | \quad \text{BH-FDR } p = 0.766$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..5) b_j \cdot x_{t-j})$

controls: ADL self-lags p=0 (BIC) ; cause block K=5 ; sample spans 4 days (probit: no day-FE) ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 4.88, \quad p = 0.43 \quad | \quad \text{BH-FDR } p = 0.762$

=> event verdict: neither direction significant.

3. OVERALL VERDICT

No Granger causality detected on either axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

=====

Trump 281 EV x VGT [ticker: KXECDJT281]

Contract : "Will Trump win 281-257 - AZ, GA, NC, PA?"

Window : 2024-11-04 to 2024-11-25 primary bar : 5min Kalshi trades : 308

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 ⇒ that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:

$y_t = \alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \sum(j=1..5) b_j \cdot x_{t-j} + \sum(d=1..10) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=2 (BIC) ; cause block K=5 ; day-FE over 11 days ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 3.36, \quad p = 0.645 \quad | \quad \text{BH-FDR } p = 0.727$

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \sum(j=1..5) d_j \cdot y_{t-j} + \sum(d=1..10) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=2 (BIC) ; cause block K=5 ; day-FE over 11 days ; hac-panel SE (bw=5).

H0: $d_1 = \dots = d_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 3.42, \quad p = 0.635 \quad | \quad \text{BH-FDR } p = 0.779$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..5) b_j \cdot x_{t-j})$

controls: ADL self-lags p=1 (BIC) ; cause block K=5 ; sample spans 11 days (probit: no day-FE) ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 5.51, \quad p = 0.357 \quad | \quad \text{BH-FDR } p = 0.586$

=> calendar verdict: neither direction significant.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi → ETF:

$y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..5) b_j \cdot x_{t-j} + \sum(d=1..3) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=0 (BIC) ; cause block K=5 ; day-FE over 4 days ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 2.47, \quad p = 0.781 \quad | \quad \text{BH-FDR } p = 0.932$

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \sum(j=1..5) d_j \cdot y_{t-j} + \sum(d=1..3) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=2 (BIC) ; cause block K=5 ; day-FE over 4 days ; hac-panel SE (bw=5).

H0: $d_1 = \dots = d_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 3.69, \quad p = 0.594 \quad | \quad \text{BH-FDR } p = 0.766$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..5) b_j \cdot x_{t-j})$

controls: ADL self-lags p=0 (BIC) ; cause block K=5 ; sample spans 4 days (probit: no day-FE) ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 1.48, \quad p = 0.915 \quad | \quad \text{BH-FDR } p = 0.967$

=> event verdict: neither direction significant.

3. OVERALL VERDICT

No Granger causality detected on either axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

=====

Trump 281 EV x VCR [ticker: KXECDJT281]

Contract : "Will Trump win 281-257 - AZ, GA, NC, PA?"

Window : 2024-11-04 to 2024-11-25 primary bar : 5min Kalshi trades : 308

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 ⇒ that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:

$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..5) b_j \cdot x_{t-j} + \sum(d=1..10) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=1 (BIC) ; cause block K=5 ; day-FE over 11 days ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 1.58, \quad p = 0.904 \quad | \quad \text{BH-FDR } p = 0.938$

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \sum(j=1..5) d_j \cdot y_{t-j} + \sum(d=1..10) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=2 (BIC) ; cause block K=5 ; day-FE over 11 days ; hac-panel SE (bw=5).

H0: $d_1 = \dots = d_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 1.85, \quad p = 0.87 \quad | \quad \text{BH-FDR } p = 0.878$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..5) b_j \cdot x_{t-j})$

controls: ADL self-lags p=1 (BIC) ; cause block K=5 ; sample spans 11 days (probit: no day-FE) ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 0.62, \quad p = 0.987 \quad | \quad \text{BH-FDR } p = 0.987$

=> calendar verdict: neither direction significant.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi → ETF:

$y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..5) b_j \cdot x_{t-j} + \sum(d=1..3) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=0 (BIC) ; cause block K=5 ; day-FE over 4 days ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 6.08, \quad p = 0.298 \quad | \quad \text{BH-FDR } p = 0.569$

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \sum(j=1..5) d_j \cdot y_{t-j} + \sum(d=1..3) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=2 (BIC) ; cause block K=5 ; day-FE over 4 days ; hac-panel SE (bw=5).

H0: $d_1 = \dots = d_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 9.37, \quad p = 0.0953 * \quad | \quad \text{BH-FDR } p = 0.222$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..5) b_j \cdot x_{t-j})$

controls: ADL self-lags p=0 (BIC) ; cause block K=5 ; sample spans 4 days (probit: no day-FE) ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 4.09, \quad p = 0.537 \quad | \quad \text{BH-FDR } p = 0.811$

=> event verdict: neither direction significant.

3. OVERALL VERDICT

No Granger causality detected on either axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

=====

Trump 312 EV x VAW [ticker: KXECDJT312]

Contract : "Will Trump win 312-226 - swing state sweep?"

Window : 2024-11-04 to 2024-12-12 primary bar : 10min Kalshi trades : 232

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).

p<0.05 ⇒ that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:

$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..5) b_j \cdot x_{t-j} + \sum(d=1..17) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=1 (BIC) ; cause block K=5 ; day-FE over 18 days ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 0.91, \quad p = 0.969 \quad | \quad \text{BH-FDR } p = 0.969$

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum(j=1..5) d_j \cdot y_{t-j} + \sum(d=1..17) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=1 (BIC) ; cause block K=5 ; day-FE over 18 days ; hac-panel SE (bw=5).

H0: $d_1 = \dots = d_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 7.42, \quad p = 0.191 \quad | \quad \text{BH-FDR } p = 0.706$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..5) b_j \cdot x_{t-j})$

controls: ADL self-lags p=0 (BIC) ; cause block K=5 ; sample spans 18 days (probit: no day-FE) ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 2.24, \quad p = 0.816 \quad | \quad \text{BH-FDR } p = 0.881$

=> calendar verdict: neither direction significant.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi → ETF:

$y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..5) b_j \cdot x_{t-j} + \sum(d=1..3) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=0 (BIC) ; cause block K=5 ; day-FE over 4 days ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 48.20, \quad p = 3.23\text{e-}09 \text{ ***} \quad | \quad \text{BH-FDR } p = 3.39\text{e-}08 \text{ ***}$

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \sum(j=1..5) d_j \cdot y_{t-j} + \sum(d=1..3) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=2 (BIC) ; cause block K=5 ; day-FE over 4 days ; hac-panel SE (bw=5).

H0: $d_1 = \dots = d_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 3.25, \quad p = 0.661 \quad | \quad \text{BH-FDR } p = 0.766$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..5) b_j \cdot x_{t-j})$

controls: ADL self-lags p=1 (BIC) ; cause block K=5 ; sample spans 4 days (probit: no day-FE) ; hac-panel SE (bw=5).

H0: $b_1 = \dots = b_5 = 0 \rightarrow \text{Wald } \chi^2(5) = 23.80, \quad p = 0.000237 \text{ ***} \quad | \quad \text{BH-FDR } p = 0.00273 \text{ ***}$

=> event verdict: Kalshi Granger-leads ETF.

3. OVERALL VERDICT

Calendar says 'neither direction significant' but Event says 'Kalshi Granger-leads ETF' — not robust across time-axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

=====

Trump 306 EV x VDC [ticker: KXECDJT306]

Contract : "Will Trump win 306-232 - AZ, GA, MI, PA, WI, NC?"

Window : 2024-11-04 to 2024-12-16 primary bar : 10min Kalshi trades : 123

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 ⇒ that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:

$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..4) b_j \cdot x_{t-j} + \sum(d=1..12) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=1 (BIC) ; cause block K=4 ; day-FE over 13 days ; hac-panel SE (bw=4).

H0: $b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 10.63, \quad p = 0.031^{**} \quad | \quad \text{BH-FDR } p = 0.105$

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \sum(j=1..4) d_j \cdot y_{t-j} + \sum(d=1..12) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=3 (BIC) ; cause block K=4 ; day-FE over 13 days ; hac-panel SE (bw=4).

H0: $d_1 = \dots = d_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 4.51, \quad p = 0.342 \quad | \quad \text{BH-FDR } p = 0.706$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..4) b_j \cdot x_{t-j})$

controls: ADL self-lags p=1 (BIC) ; cause block K=4 ; sample spans 13 days (probit: no day-FE) ; hac-panel SE (bw=4).

H0: $b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 3.80, \quad p = 0.434 \quad | \quad \text{BH-FDR } p = 0.586$

⇒ calendar verdict: Kalshi Granger-leads ETF.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi → ETF:

$y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..4) b_j \cdot x_{t-j} + \sum(d=1..1) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=0 (BIC) ; cause block K=4 ; day-FE over 2 days ; hac-panel SE (bw=4).

H0: $b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 51.18, \quad p = 2.05\text{e-}10^{***} \quad | \quad \text{BH-FDR } p = 4.31\text{e-}09^{***}$

Direction B — ETF → Kalshi:

$x_t = \alpha + (\text{no } x \text{ self-lag term: BIC chose } p=0) + \sum(j=1..4) d_j \cdot y_{t-j} + \sum(d=1..1) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=0 (BIC) ; cause block K=4 ; day-FE over 2 days ; hac-panel SE (bw=4).

H0: $d_1 = \dots = d_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 7.35, \quad p = 0.119 \quad | \quad \text{BH-FDR } p = 0.249$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..4) b_j \cdot x_{t-j})$

controls: ADL self-lags p=0 (BIC) ; cause block K=4 ; sample spans 2 days (probit: no day-FE) ; hac-panel SE (bw=4).

H0: $b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 8.22, \quad p = 0.0839^* \quad | \quad \text{BH-FDR } p = 0.276$

⇒ event verdict: Kalshi Granger-leads ETF.

3. OVERALL VERDICT

Both time-axes agree: Kalshi Granger-leads ETF (relatively robust).

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

=====

Trump 316 EV x VFH [ticker: KXECDJT316]

Contract : "Will Trump win 316-222 - AZ, GA, MI, MN, NC, PA, WI?"

Window : 2024-11-04 to 2024-12-12 primary bar : 10min Kalshi trades : 59

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 ⇒ that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:

$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j} + \sum_{d=1..6} \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=1 (BIC) ; cause block K=4 ; day-FE over 7 days ; hac-panel SE (bw=4).

H0: $b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 4.24, \quad p = 0.374 \quad | \quad \text{BH-FDR } p = 0.594$

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum_{j=1..4} d_j \cdot y_{t-j} + \sum_{d=1..6} \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=1 (BIC) ; cause block K=4 ; day-FE over 7 days ; hac-panel SE (bw=4).

H0: $d_1 = \dots = d_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 5.67, \quad p = 0.225 \quad | \quad \text{BH-FDR } p = 0.706$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \sum_{j=1..4} b_j \cdot x_{t-j})$

controls: ADL self-lags p=2 (BIC) ; cause block K=4 ; sample spans 7 days (probit: no day-FE) ; hac-panel SE (bw=4).

H0: $b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 5.48, \quad p = 0.242 \quad | \quad \text{BH-FDR } p = 0.586$

⇒ calendar verdict: neither direction significant.

2. EVENT-TIME (event-count lags)

(not estimable at this axis)

3. OVERALL VERDICT

No Granger causality detected on either axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

=====

Trump 316 EV x VIS [ticker: KXECDJT316]

Contract : "Will Trump win 316-222 - AZ, GA, MI, MN, NC, PA, WI?"

Window : 2024-11-04 to 2024-12-12 primary bar : 10min Kalshi trades : 59

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 ⇒ that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:

$y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..4) b_j \cdot x_{t-j} + \sum(d=1..6) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=0 (BIC) ; cause block K=4 ; day-FE over 7 days ; hac-panel SE (bw=4).

H0: $b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 6.45, \quad p = 0.168 \quad | \quad \text{BH-FDR } p = 0.324$

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum(j=1..4) d_j \cdot y_{t-j} + \sum(d=1..6) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=1 (BIC) ; cause block K=4 ; day-FE over 7 days ; hac-panel SE (bw=4).

H0: $d_1 = \dots = d_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 6.09, \quad p = 0.192 \quad | \quad \text{BH-FDR } p = 0.706$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \sum(j=1..4) b_j \cdot x_{t-j})$

controls: ADL self-lags p=2 (BIC) ; cause block K=4 ; sample spans 7 days (probit: no day-FE) ; hac-panel SE (bw=4).

H0: $b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 10.05, \quad p = 0.0395 \text{ **} \quad | \quad \text{BH-FDR } p = 0.213$

⇒ calendar verdict: neither direction significant.

2. EVENT-TIME (event-count lags)

(not estimable at this axis)

3. OVERALL VERDICT

No Granger causality detected on either axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

=====

Trump 316 EV x VDE [ticker: KXECDJT316]

Contract : "Will Trump win 316-222 - AZ, GA, MI, MN, NC, PA, WI?"

Window : 2024-11-04 to 2024-12-12 primary bar : 10min Kalshi trades : 59

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 ⇒ that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:

$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j} + \sum_{d=1..6} \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=1 (BIC) ; cause block K=4 ; day-FE over 7 days ; hac-panel SE (bw=4).

H0: $b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 14.19, \quad p = 0.00672 \text{ ***} \quad | \quad \text{BH-FDR } p = 0.0336 \text{ **}$

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum_{j=1..4} d_j \cdot y_{t-j} + \sum_{d=1..6} \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=1 (BIC) ; cause block K=4 ; day-FE over 7 days ; hac-panel SE (bw=4).

H0: $d_1 = \dots = d_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 3.43, \quad p = 0.489 \quad | \quad \text{BH-FDR } p = 0.741$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j})$

controls: ADL self-lags p=1 (BIC) ; cause block K=4 ; sample spans 7 days (probit: no day-FE) ; hac-panel SE (bw=4).

H0: $b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 7.22, \quad p = 0.125 \quad | \quad \text{BH-FDR } p = 0.429$

⇒ calendar verdict: Kalshi Granger-leads ETF.

2. EVENT-TIME (event-count lags)

(not estimable at this axis)

3. OVERALL VERDICT

Only one axis significant: Kalshi Granger-leads ETF.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

=====

Gas >\$3.15 by Oct31 x VPU [ticker: AAAGASM-240CT31-US-3.15]

Contract : "Will average **gas prices** be above \$3.15?"

Window : 2024-10-02 to 2024-10-31 primary bar : 10min Kalshi trades : 58

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 ⇒ that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:

$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..3} b_j \cdot x_{t-j} + \sum_{d=1..13} \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=1 (BIC) ; cause block K=3 ; day-FE over 14 days ; hac-panel SE (bw=3).

H0: $b_1 = \dots = b_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 15.13, \quad p = 0.00171 \text{ ***} \quad | \quad \text{BH-FDR } p = 0.0116 \text{ **}$

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \sum_{j=1..3} d_j \cdot y_{t-j} + \sum_{d=1..13} \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=2 (BIC) ; cause block K=3 ; day-FE over 14 days ; hac-panel SE (bw=3).

H0: $d_1 = \dots = d_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 1.97, \quad p = 0.578 \quad | \quad \text{BH-FDR } p = 0.779$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..3} b_j \cdot x_{t-j})$

controls: ADL self-lags p=1 (BIC) ; cause block K=3 ; sample spans 14 days (probit: no day-FE) ; hac-panel SE (bw=3).

H0: $b_1 = \dots = b_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 6.87, \quad p = 0.076 \text{ * } \quad | \quad \text{BH-FDR } p = 0.342$

⇒ calendar verdict: Kalshi Granger-leads ETF.

2. EVENT-TIME (event-count lags)

(not estimable at this axis)

3. OVERALL VERDICT

Only one axis significant: Kalshi Granger-leads ETF.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

=====

Harris 276 EV x VOX [ticker: KXECKH276]

Contract : "Will Harris win 276-262 - PA, NV, MI, WI?"

Window : 2024-11-04 to 2024-11-26 primary bar : 10min Kalshi trades : 43

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 ⇒ that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:

$y_t = \alpha + \varphi_1 \cdot y_{t-1} + \sum(j=1..3) b_j \cdot x_{t-j} + \sum(d=1..5) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=1 (BIC) ; cause block K=3 ; day-FE over 6 days ; hac-panel SE (bw=3).

H0: $b_1 = \dots = b_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 11.98, \quad p = 0.00746 \text{ ***} \quad | \quad \text{BH-FDR } p = 0.0336 \text{ **}$

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum(j=1..3) d_j \cdot y_{t-j} + \sum(d=1..5) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=1 (BIC) ; cause block K=3 ; day-FE over 6 days ; hac-panel SE (bw=3).

H0: $d_1 = \dots = d_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 4.26, \quad p = 0.235 \quad | \quad \text{BH-FDR } p = 0.706$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..3) b_j \cdot x_{t-j})$

controls: ADL self-lags p=0 (BIC) ; cause block K=3 ; sample spans 6 days (probit: no day-FE) ; hac-panel SE (bw=3).

H0: $b_1 = \dots = b_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 9.68, \quad p = 0.0215 \text{ **} \quad | \quad \text{BH-FDR } p = 0.193$

=> calendar verdict: Kalshi Granger-leads ETF.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi → ETF:

$y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..3) b_j \cdot x_{t-j}$

controls: ADL self-lags p=0 (BIC) ; cause block K=3 ; no day-FE (single day) ; hac-panel SE (bw=3).

H0: $b_1 = \dots = b_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 3.10, \quad p = 0.376 \quad | \quad \text{BH-FDR } p = 0.607$

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \sum(j=1..3) d_j \cdot y_{t-j}$

controls: ADL self-lags p=3 (BIC) ; cause block K=3 ; no day-FE (single day) ; hac-panel SE (bw=3).

H0: $d_1 = \dots = d_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 0.74, \quad p = 0.863 \quad | \quad \text{BH-FDR } p = 0.863$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..3) b_j \cdot x_{t-j})$

controls: ADL self-lags p=0 (BIC) ; cause block K=3 ; sample spans 1 days (probit: no day-FE) ; hac-panel SE (bw=3).

H0: $b_1 = \dots = b_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 3.36, \quad p = 0.34 \quad | \quad \text{BH-FDR } p = 0.652$

=> event verdict: neither direction significant.

3. OVERALL VERDICT

Calendar says 'Kalshi Granger-leads ETF' but Event says 'neither direction significant' — not robust across time-axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

Harris 276 EV x VFH [ticker: KXECKH276]
Contract : "Will Harris win 276-262 - PA, NV, MI, WI?"
Window : 2024-11-04 to 2024-11-26 primary bar : 10min Kalshi trades : 43

Test: joint Wald H_0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
 $p < 0.05 \Rightarrow$ that variable Granger-leads the other. (x= Δ prob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi $\rightarrow$ ETF:
 $y_t = \alpha +$ (no y self-lag term: BIC chose $p=0$) $+ \sum(j=1..3) b_j \cdot x_{t-j} + \sum(d=1..5) \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags $p=0$ (BIC) ; cause block $K=3$; day-FE over 6 days ; hac-panel SE (bw=3).
 $H_0: b_1 = \dots = b_3 = 0 \rightarrow$ Wald $\chi^2(3) = 20.56, p = 0.00013 ***$ | BH-FDR $p = 0.00117 ***$
Direction B — ETF $\rightarrow$ Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum(j=1..3) d_j \cdot y_{t-j} + \sum(d=1..5) \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags $p=1$ (BIC) ; cause block $K=3$; day-FE over 6 days ; hac-panel SE (bw=3).
 $H_0: d_1 = \dots = d_3 = 0 \rightarrow$ Wald $\chi^2(3) = 3.52, p = 0.319$ | BH-FDR $p = 0.706$
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha +$ (no y self-lag term: BIC chose $p=0$) $+ \sum(j=1..3) b_j \cdot x_{t-j})$
controls: ADL self-lags $p=0$ (BIC) ; cause block $K=3$; sample spans 6 days (probit: no day-FE) ; hac-panel SE (bw=3).
 $H_0: b_1 = \dots = b_3 = 0 \rightarrow$ Wald $\chi^2(3) = 0.17, p = 0.982$ | BH-FDR $p = 0.987$
 $\Rightarrow$ calendar verdict: Kalshi Granger-leads ETF.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi $\rightarrow$ ETF:
 $y_t = \alpha +$ (no y self-lag term: BIC chose $p=0$) $+ \sum(j=1..3) b_j \cdot x_{t-j}$
controls: ADL self-lags $p=0$ (BIC) ; cause block $K=3$; no day-FE (single day) ; hac-panel SE (bw=3).
 $H_0: b_1 = \dots = b_3 = 0 \rightarrow$ Wald $\chi^2(3) = 1.62, p = 0.655$ | BH-FDR $p = 0.86$
Direction B — ETF $\rightarrow$ Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \sum(j=1..3) d_j \cdot y_{t-j}$
controls: ADL self-lags $p=3$ (BIC) ; cause block $K=3$; no day-FE (single day) ; hac-panel SE (bw=3).
 $H_0: d_1 = \dots = d_3 = 0 \rightarrow$ Wald $\chi^2(3) = 7.19, p = 0.0659 *$ | BH-FDR $p = 0.173$
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha +$ (no y self-lag term: BIC chose $p=0$) $+ \sum(j=1..3) b_j \cdot x_{t-j})$
controls: ADL self-lags $p=0$ (BIC) ; cause block $K=3$; sample spans 1 days (probit: no day-FE) ; hac-panel SE (bw=3).
 $H_0: b_1 = \dots = b_3 = 0 \rightarrow$ Wald $\chi^2(3) = 0.26, p = 0.967$ | BH-FDR $p = 0.967$
 $\Rightarrow$ event verdict: neither direction significant.

3. OVERALL VERDICT

Calendar says 'Kalshi Granger-leads ETF' but Event says 'neither direction significant' — not robust across time-axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

=====

Harris 276 EV x VIS [ticker: KXECKH276]

Contract : "Will Harris win 276-262 - PA, NV, MI, WI?"

Window : 2024-11-04 to 2024-11-26 primary bar : 10min Kalshi trades : 43

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 ⇒ that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:

$y_t = \alpha +$ (no y self-lag term: BIC chose p=0) $+ \sum(j=1..3) b_j \cdot x_{t-j} + \sum(d=1..5) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=0 (BIC) ; cause block K=3 ; day-FE over 6 days ; hac-panel SE (bw=3).

H0: $b_1 = \dots = b_3 = 0 \rightarrow$ Wald $\chi^2(3) = 22.17$, p = 6.01e-05 *** | BH-FDR p = 0.000812 ***

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum(j=1..3) d_j \cdot y_{t-j} + \sum(d=1..5) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags p=1 (BIC) ; cause block K=3 ; day-FE over 6 days ; hac-panel SE (bw=3).

H0: $d_1 = \dots = d_3 = 0 \rightarrow$ Wald $\chi^2(3) = 3.59$, p = 0.309 | BH-FDR p = 0.706

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha +$ (no y self-lag term: BIC chose p=0) $+ \sum(j=1..3) b_j \cdot x_{t-j})$

controls: ADL self-lags p=0 (BIC) ; cause block K=3 ; sample spans 6 days (probit: no day-FE) ; hac-panel SE (bw=3).

H0: $b_1 = \dots = b_3 = 0 \rightarrow$ Wald $\chi^2(3) = 23.93$, p = 2.59e-05 *** | BH-FDR p = 0.000699 ***

=> calendar verdict: Kalshi Granger-leads ETF.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi → ETF:

$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..3) b_j \cdot x_{t-j}$

controls: ADL self-lags p=1 (BIC) ; cause block K=3 ; no day-FE (single day) ; hac-panel SE (bw=3).

H0: $b_1 = \dots = b_3 = 0 \rightarrow$ Wald $\chi^2(3) = 6.16$, p = 0.104 | BH-FDR p = 0.219

Direction B — ETF → Kalshi:

$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \sum(j=1..3) d_j \cdot y_{t-j}$

controls: ADL self-lags p=3 (BIC) ; cause block K=3 ; no day-FE (single day) ; hac-panel SE (bw=3).

H0: $d_1 = \dots = d_3 = 0 \rightarrow$ Wald $\chi^2(3) = 12.19$, p = 0.00676 *** | BH-FDR p = 0.0355 **

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha +$ (no y self-lag term: BIC chose p=0) $+ \sum(j=1..3) b_j \cdot x_{t-j})$

controls: ADL self-lags p=0 (BIC) ; cause block K=3 ; sample spans 1 days (probit: no day-FE) ; hac-panel SE (bw=3).

H0: $b_1 = \dots = b_3 = 0 \rightarrow$ Wald $\chi^2(3) = 4.45$, p = 0.217 | BH-FDR p = 0.472

=> event verdict: ETF Granger-leads Kalshi.

3. OVERALL VERDICT

Calendar says 'Kalshi Granger-leads ETF' but Event says 'ETF Granger-leads Kalshi' — not robust across time-axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

Harris 276 EV x VCR [ticker: KXECKH276]
Contract : "Will Harris win 276-262 - PA, NV, MI, WI?"
Window : 2024-11-04 to 2024-11-26 primary bar : 10min Kalshi trades : 43

Test: joint Wald H_0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
 $p < 0.05 \Rightarrow$ that variable Granger-leads the other. (x= Δ prob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi $\rightarrow$ ETF:
 $y_t = \alpha +$ (no y self-lag term: BIC chose $p=0$) $+ \sum(j=1..3) b_j \cdot x_{t-j} + \sum(d=1..5) \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags $p=0$ (BIC) ; cause block $K=3$; day-FE over 6 days ; hac-panel SE (bw=3).
 $H_0: b_1 = \dots = b_3 = 0 \rightarrow$ Wald $\chi^2(3) = 7.20, p = 0.0658$ | BH-FDR $p = 0.197$
Direction B — ETF $\rightarrow$ Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum(j=1..3) d_j \cdot y_{t-j} + \sum(d=1..5) \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags $p=1$ (BIC) ; cause block $K=3$; day-FE over 6 days ; hac-panel SE (bw=3).
 $H_0: d_1 = \dots = d_3 = 0 \rightarrow$ Wald $\chi^2(3) = 2.83, p = 0.418$ | BH-FDR $p = 0.706$
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha +$ (no y self-lag term: BIC chose $p=0$) $+ \sum(j=1..3) b_j \cdot x_{t-j})$
controls: ADL self-lags $p=0$ (BIC) ; cause block $K=3$; sample spans 6 days (probit: no day-FE) ; hac-panel SE (bw=3).
 $H_0: b_1 = \dots = b_3 = 0 \rightarrow$ Wald $\chi^2(3) = 4.20, p = 0.24$ | BH-FDR $p = 0.586$
 $\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi $\rightarrow$ ETF:
 $y_t = \alpha +$ (no y self-lag term: BIC chose $p=0$) $+ \sum(j=1..3) b_j \cdot x_{t-j}$
controls: ADL self-lags $p=0$ (BIC) ; cause block $K=3$; no day-FE (single day) ; hac-panel SE (bw=3).
 $H_0: b_1 = \dots = b_3 = 0 \rightarrow$ Wald $\chi^2(3) = 1.91, p = 0.591$ | BH-FDR $p = 0.836$
Direction B — ETF $\rightarrow$ Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \sum(j=1..3) d_j \cdot y_{t-j}$
controls: ADL self-lags $p=3$ (BIC) ; cause block $K=3$; no day-FE (single day) ; hac-panel SE (bw=3).
 $H_0: d_1 = \dots = d_3 = 0 \rightarrow$ Wald $\chi^2(3) = 39.91, p = 1.11e-08$ *** | BH-FDR $p = 1.04e-07$ ***
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha +$ (no y self-lag term: BIC chose $p=0$) $+ \sum(j=1..3) b_j \cdot x_{t-j})$
controls: ADL self-lags $p=0$ (BIC) ; cause block $K=3$; sample spans 1 days (probit: no day-FE) ; hac-panel SE (bw=3).
 $H_0: b_1 = \dots = b_3 = 0 \rightarrow$ Wald $\chi^2(3) = 1.09, p = 0.778$ | BH-FDR $p = 0.895$
 $\Rightarrow$ event verdict: ETF Granger-leads Kalshi.

3. OVERALL VERDICT

Calendar says 'neither direction significant' but Event says 'ETF Granger-leads Kalshi' — not robust across time-axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

Harris 276 EV x VDE [ticker: KXECKH276]
Contract : "Will Harris win 276-262 - PA, NV, MI, WI?"
Window : 2024-11-04 to 2024-11-26 primary bar : 10min Kalshi trades : 43

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 ⇒ that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:
 $y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..3) b_j \cdot x_{t-j} + \sum(d=1..5) \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=1 (BIC) ; cause block K=3 ; day-FE over 6 days ; hac-panel SE (bw=3).
H0: $b_1 = \dots = b_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 33.54, p = 2.48\text{e-}07 \text{ ***} \mid \text{BH-FDR } p = 6.69\text{e-}06 \text{ ***}$
Direction B — ETF → Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum(j=1..3) d_j \cdot y_{t-j} + \sum(d=1..5) \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=1 (BIC) ; cause block K=3 ; day-FE over 6 days ; hac-panel SE (bw=3).
H0: $d_1 = \dots = d_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 2.86, p = 0.414 \mid \text{BH-FDR } p = 0.706$
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..3) b_j \cdot x_{t-j})$
controls: ADL self-lags p=1 (BIC) ; cause block K=3 ; sample spans 6 days (probit: no day-FE) ; hac-panel SE (bw=3).
H0: $b_1 = \dots = b_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 8.51, p = 0.0366 \text{ **} \mid \text{BH-FDR } p = 0.213$
=> calendar verdict: Kalshi Granger-leads ETF.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi → ETF:
 $y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..3) b_j \cdot x_{t-j}$
controls: ADL self-lags p=0 (BIC) ; cause block K=3 ; no day-FE (single day) ; hac-panel SE (bw=3).
H0: $b_1 = \dots = b_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 13.36, p = 0.00391 \text{ ***} \mid \text{BH-FDR } p = 0.0164 \text{ **}$
Direction B — ETF → Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \sum(j=1..3) d_j \cdot y_{t-j}$
controls: ADL self-lags p=3 (BIC) ; cause block K=3 ; no day-FE (single day) ; hac-panel SE (bw=3).
H0: $d_1 = \dots = d_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 9.47, p = 0.0237 \text{ **} \mid \text{BH-FDR } p = 0.0994 \text{ *}$
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..3) b_j \cdot x_{t-j})$
controls: ADL self-lags p=1 (BIC) ; cause block K=3 ; sample spans 1 days (probit: no day-FE) ; hac-panel SE (bw=3).
H0: $b_1 = \dots = b_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 4.59, p = 0.205 \mid \text{BH-FDR } p = 0.472$
=> event verdict: bidirectional (both reject).

3. OVERALL VERDICT

Calendar says 'Kalshi Granger-leads ETF' but Event says 'bidirectional (both reject)' — not robust across time-axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

Harris 276 EV x VGT [ticker: KXECKH276]
Contract : "Will Harris win 276-262 - PA, NV, MI, WI?"
Window : 2024-11-04 to 2024-11-26 primary bar : 10min Kalshi trades : 43

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 => that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:
y_t = α + (no y self-lag term: BIC chose p=0) + Σ(j=1..3) b_j·x_{t-j} + Σ(d=1..5) γ_d·Day_d
controls: ADL self-lags p=0 (BIC) ; cause block K=3 ; day-FE over 6 days ; hac-panel SE (bw=3).
H0: b₁ = ... = b₃ = 0 → Wald χ²(3) = 2.20, p = 0.533 | BH-FDR p = 0.685
Direction B — ETF → Kalshi:
x_t = α + ψ₁·x_{t-1} + Σ(j=1..3) d_j·y_{t-j} + Σ(d=1..5) γ_d·Day_d
controls: ADL self-lags p=1 (BIC) ; cause block K=3 ; day-FE over 6 days ; hac-panel SE (bw=3).
H0: d₁ = ... = d₃ = 0 → Wald χ²(3) = 3.27, p = 0.352 | BH-FDR p = 0.706
Direction A (probit, ETF up/down):
Pr(y_t>0) = Φ(α + (no y self-lag term: BIC chose p=0) + Σ(j=1..3) b_j·x_{t-j})
controls: ADL self-lags p=0 (BIC) ; cause block K=3 ; sample spans 6 days (probit: no day-FE) ; hac-panel SE (bw=3).
H0: b₁ = ... = b₃ = 0 → Wald χ²(3) = 1.34, p = 0.72 | BH-FDR p = 0.81
=> calendar verdict: neither direction significant.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi → ETF:
y_t = α + (no y self-lag term: BIC chose p=0) + Σ(j=1..3) b_j·x_{t-j}
controls: ADL self-lags p=0 (BIC) ; cause block K=3 ; no day-FE (single day) ; hac-panel SE (bw=3).
H0: b₁ = ... = b₃ = 0 → Wald χ²(3) = 36.21, p = 6.75e-08 *** | BH-FDR p = 4.72e-07 ***
Direction B — ETF → Kalshi:
x_t = α + ψ₁·x_{t-1} + ψ₂·x_{t-2} + ψ₃·x_{t-3} + Σ(j=1..3) d_j·y_{t-j}
controls: ADL self-lags p=3 (BIC) ; cause block K=3 ; no day-FE (single day) ; hac-panel SE (bw=3).
H0: d₁ = ... = d₃ = 0 → Wald χ²(3) = 8.50, p = 0.0367 ** | BH-FDR p = 0.11
Direction A (probit, ETF up/down):
Pr(y_t>0) = Φ(α + (no y self-lag term: BIC chose p=0) + Σ(j=1..3) b_j·x_{t-j})
controls: ADL self-lags p=0 (BIC) ; cause block K=3 ; sample spans 1 days (probit: no day-FE) ; hac-panel SE (bw=3).
H0: b₁ = ... = b₃ = 0 → Wald χ²(3) = 4.36, p = 0.226 | BH-FDR p = 0.472
=> event verdict: bidirectional (both reject).

3. OVERALL VERDICT

Calendar says 'neither direction significant' but Event says 'bidirectional (both reject)' — not robust across time-axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

Harris 276 EV x VDC [ticker: KXECKH276]
Contract : "Will Harris win 276-262 - PA, NV, MI, WI?"
Window : 2024-11-04 to 2024-11-26 primary bar : 10min Kalshi trades : 43

Test: joint Wald H_0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
 $p < 0.05 \Rightarrow$ that variable Granger-leads the other. (x= Δ prob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi $\rightarrow$ ETF:
 $y_t = \alpha +$ (no y self-lag term: BIC chose $p=0$) $+ \sum(j=1..3) b_j \cdot x_{t-j} + \sum(d=1..5) \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags $p=0$ (BIC) ; cause block $K=3$; day-FE over 6 days ; hac-panel SE (bw=3).
 $H_0: b_1 = \dots = b_3 = 0 \rightarrow$ Wald $\chi^2(3) = 3.55, p = 0.314$ | BH-FDR $p = 0.531$
Direction B — ETF $\rightarrow$ Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum(j=1..3) d_j \cdot y_{t-j} + \sum(d=1..5) \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags $p=1$ (BIC) ; cause block $K=3$; day-FE over 6 days ; hac-panel SE (bw=3).
 $H_0: d_1 = \dots = d_3 = 0 \rightarrow$ Wald $\chi^2(3) = 1.72, p = 0.633$ | BH-FDR $p = 0.779$
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha +$ (no y self-lag term: BIC chose $p=0$) $+ \sum(j=1..3) b_j \cdot x_{t-j})$
controls: ADL self-lags $p=0$ (BIC) ; cause block $K=3$; sample spans 6 days (probit: no day-FE) ; hac-panel SE (bw=3).
 $H_0: b_1 = \dots = b_3 = 0 \rightarrow$ Wald $\chi^2(3) = 3.64, p = 0.304$ | BH-FDR $p = 0.586$
 $\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi $\rightarrow$ ETF:
 $y_t = \alpha +$ (no y self-lag term: BIC chose $p=0$) $+ \sum(j=1..3) b_j \cdot x_{t-j}$
controls: ADL self-lags $p=0$ (BIC) ; cause block $K=3$; no day-FE (single day) ; hac-panel SE (bw=3).
 $H_0: b_1 = \dots = b_3 = 0 \rightarrow$ Wald $\chi^2(3) = 0.82, p = 0.844$ | BH-FDR $p = 0.932$
Direction B — ETF $\rightarrow$ Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \sum(j=1..3) d_j \cdot y_{t-j}$
controls: ADL self-lags $p=3$ (BIC) ; cause block $K=3$; no day-FE (single day) ; hac-panel SE (bw=3).
 $H_0: d_1 = \dots = d_3 = 0 \rightarrow$ Wald $\chi^2(3) = 39.33, p = 1.48e-08$ *** | BH-FDR $p = 1.04e-07$ ***
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha +$ (no y self-lag term: BIC chose $p=0$) $+ \sum(j=1..3) b_j \cdot x_{t-j})$
controls: ADL self-lags $p=0$ (BIC) ; cause block $K=3$; sample spans 1 days (probit: no day-FE) ; hac-panel SE (bw=3).
 $H_0: b_1 = \dots = b_3 = 0 \rightarrow$ Wald $\chi^2(3) = 2.04, p = 0.565$ | BH-FDR $p = 0.811$
 $\Rightarrow$ event verdict: ETF Granger-leads Kalshi.

3. OVERALL VERDICT

Calendar says 'neither direction significant' but Event says 'ETF Granger-leads Kalshi' — not robust across time-axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

Harris 276 EV x VAW [ticker: KXECKH276]
Contract : "Will Harris win 276-262 - PA, NV, MI, WI?"
Window : 2024-11-04 to 2024-11-26 primary bar : 10min Kalshi trades : 43

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 => that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:
 $y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..3) b_j \cdot x_{t-j} + \sum(d=1..5) \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=1 (BIC) ; cause block K=3 ; day-FE over 6 days ; hac-panel SE (bw=3).
H0: $b_1 = \dots = b_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 11.60, p = 0.00889 \text{ ***} \mid \text{BH-FDR } p = 0.0343 \text{ **}$
Direction B — ETF → Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum(j=1..3) d_j \cdot y_{t-j} + \sum(d=1..5) \gamma_d \cdot \text{Day}_d$
controls: ADL self-lags p=1 (BIC) ; cause block K=3 ; day-FE over 6 days ; hac-panel SE (bw=3).
H0: $d_1 = \dots = d_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 3.37, p = 0.338 \mid \text{BH-FDR } p = 0.706$
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..3) b_j \cdot x_{t-j})$
controls: ADL self-lags p=0 (BIC) ; cause block K=3 ; sample spans 6 days (probit: no day-FE) ; hac-panel SE (bw=3).
H0: $b_1 = \dots = b_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 2.91, p = 0.406 \mid \text{BH-FDR } p = 0.586$
=> calendar verdict: Kalshi Granger-leads ETF.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi → ETF:
 $y_t = \alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \phi_3 \cdot y_{t-3} + \sum(j=1..3) b_j \cdot x_{t-j}$
controls: ADL self-lags p=3 (BIC) ; cause block K=3 ; no day-FE (single day) ; hac-panel SE (bw=3).
H0: $b_1 = \dots = b_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 8.00, p = 0.046 \text{ **} \mid \text{BH-FDR } p = 0.107$
Direction B — ETF → Kalshi:
 $x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \sum(j=1..3) d_j \cdot y_{t-j}$
controls: ADL self-lags p=3 (BIC) ; cause block K=3 ; no day-FE (single day) ; hac-panel SE (bw=3).
H0: $d_1 = \dots = d_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 8.75, p = 0.0328 \text{ **} \mid \text{BH-FDR } p = 0.11$
Direction A (probit, ETF up/down):
 $\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \phi_3 \cdot y_{t-3} + \sum(j=1..3) b_j \cdot x_{t-j})$
controls: ADL self-lags p=3 (BIC) ; cause block K=3 ; sample spans 1 days (probit: no day-FE) ; hac-panel SE (bw=3).
H0: $b_1 = \dots = b_3 = 0 \rightarrow \text{Wald } \chi^2(3) = 67.80, p = 1.26\text{e-}14 \text{ ***} \mid \text{BH-FDR } p = 2.9\text{e-}13 \text{ ***}$
=> event verdict: bidirectional (both reject).

3. OVERALL VERDICT

Calendar says 'Kalshi Granger-leads ETF' but Event says 'bidirectional (both reject)' — not robust across time-axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

Harris 287 EV x VNQ [ticker: KXECKH287]
Contract : "Will Harris win 287-251 - PA, NV, MI, WI, AZ?"
Window : 2024-11-04 to 2024-11-21 primary bar : 10min Kalshi trades : 39

Test: joint Wald H0 = all PAST cause-lags are 0 (controls the effect's own past; hac-panel SE).
p<0.05 => that variable Granger-leads the other. (x=Δprob, y=ETF log return.)

1. CALENDAR-TIME (clock-time lags, full RTH grid)

Direction A — Kalshi → ETF:
y_t = α + (no y self-lag term: BIC chose p=0) + Σ(j=1..3) b_j·x_{t-j} + Σ(d=1..4) γ_d·Day_d
controls: ADL self-lags p=0 (BIC) ; cause block K=3 ; day-FE over 5 days ; hac-panel SE (bw=3).
H0: b₁ = ... = b₃ = 0 → Wald χ²(3) = 1.66, p = 0.646 | BH-FDR p = 0.727

Direction B — ETF → Kalshi:
x_t = α + ψ₁·x_{t-1} + Σ(j=1..3) d_j·y_{t-j} + Σ(d=1..4) γ_d·Day_d
controls: ADL self-lags p=1 (BIC) ; cause block K=3 ; day-FE over 5 days ; hac-panel SE (bw=3).
H0: d₁ = ... = d₃ = 0 → Wald χ²(3) = 3.45, p = 0.328 | BH-FDR p = 0.706

Direction A (probit, ETF up/down):
Pr(y_t>0) = Φ(α + (no y self-lag term: BIC chose p=0) + Σ(j=1..3) b_j·x_{t-j})
controls: ADL self-lags p=0 (BIC) ; cause block K=3 ; sample spans 5 days (probit: no day-FE) ; hac-panel SE (bw=3).
H0: b₁ = ... = b₃ = 0 → Wald χ²(3) = 5.70, p = 0.127 | BH-FDR p = 0.429
=> calendar verdict: neither direction significant.

2. EVENT-TIME (event-count lags)

Direction A — Kalshi → ETF:
y_t = α + φ₁·y_{t-1} + φ₂·y_{t-2} + φ₃·y_{t-3} + Σ(j=1..3) b_j·x_{t-j} + Σ(d=1..1) γ_d·Day_d
controls: ADL self-lags p=3 (BIC) ; cause block K=3 ; day-FE over 2 days ; hac-panel SE (bw=3).
H0: b₁ = ... = b₃ = 0 → Wald χ²(3) = 12.71, p = 0.00531 *** | BH-FDR p = 0.0186 **

Direction B — ETF → Kalshi:
x_t = α + ψ₁·x_{t-1} + ψ₂·x_{t-2} + ψ₃·x_{t-3} + Σ(j=1..3) d_j·y_{t-j} + Σ(d=1..1) γ_d·Day_d
controls: ADL self-lags p=3 (BIC) ; cause block K=3 ; day-FE over 2 days ; hac-panel SE (bw=3).
H0: d₁ = ... = d₃ = 0 → Wald χ²(3) = 41.76, p = 4.5e-09 *** | BH-FDR p = 9.46e-08 ***

Direction A (probit, ETF up/down):
Pr(y_t>0) = Φ(α + φ₁·y_{t-1} + φ₂·y_{t-2} + φ₃·y_{t-3} + Σ(j=1..3) b_j·x_{t-j})
controls: ADL self-lags p=3 (BIC) ; cause block K=3 ; sample spans 2 days (probit: no day-FE) ; hac-panel SE (bw=3).
H0: b₁ = ... = b₃ = 0 → Wald χ²(3) = 13.81, p = 0.00317 *** | BH-FDR p = 0.0225 **
=> event verdict: bidirectional (both reject).

3. OVERALL VERDICT

Calendar says 'neither direction significant' but Event says 'bidirectional (both reject)' — not robust across time-axis.

This pair's joint-Wald p per direction is also in the single-pair heatmap (core figure 1).

MERGE POOLED SIGNALS – GRANGER (per group × ETF detail)

MERGE pooled signals – Granger detail

=====

Each pooled signal treated as one combined contract; Granger vs each ETF in its union.

MERGE ELECTION_trump_fav – Granger overview

MERGE pooled signal – GRANGER overview : ELECTION_trump_fav

=====

Each pooled signal = members' sign-aligned Δ prob pooled (member fixed effects, hac-panel SE by member $\times$ day). 'Sign-aligned' means each member's Δ prob is multiplied by ± 1 so that a positive value always means the same real-world direction (e.g. a MIN-side contract is flipped). The CONTINUOUS magnitude of Δ prob is kept – the moves are NOT discretized to ± 1 .

Joint Wald per direction; S=combined pooled signal, E=ETF. $p < 0.05 \Rightarrow$ that side leads.

ETF	cal S→E	cal E→S	evt S→E	evt E→S	verdict(cal / evt)
VAW	0.49	0.42	1.3e-05***	0.21	- / S→E
VCR	0.065*	0.37	0.086*	0.25	- / -
VDC	0.0096***	0.24	0.014**	0.19	S→E / S→E
VDE	0.13	0.2	0.0019***	0.022**	- / bidir
VFH	0.00027***	0.8	2.7e-05***	0.59	S→E / S→E
VGT	0.69	0.66	0.14	0.72	- / -
VIS	0.028**	0.35	4.9e-06***	0.24	S→E / S→E
VNQ	0.32	0.45	0.0019***	0.03**	- / bidir
VOX	0.2	0.18	0.00017***	0.34	- / S→E

Per-ETF pseudo-pairs (combined pooled signal $\times$ each ETF) follow, each with explicit formulas + joint Wald.

GRANGER – ELECTION_trump_fav (combined pooled signal) x VAW

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled – NOT a price level). $x=\Delta$ prob(S), y =ETF log return.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A – Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..6) b_j \cdot x_{t-j} + \sum(d=1..21) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=6$; day-FE over 22 days ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 5.47, \quad p = 0.485 \quad | \quad \text{BH-FDR } p = 0.561$$

Direction B – ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..6) d_j \cdot y_{t-j} + \sum(d=1..21) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=6$; day-FE over 22 days ; hac-panel SE (bw=6).

$$H_0: d_1 = \dots = d_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 6.04, \quad p = 0.418 \quad | \quad \text{BH-FDR } p = 0.735$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..6) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=6$; sample spans 22 days (probit: no day-FE) ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 8.27, \quad p = 0.219 \quad | \quad \text{BH-FDR } p = 0.476$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

Direction A – Pooled $\rightarrow$ ETF:

$$y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..6) b_j \cdot x_{t-j} + \sum(d=1..4) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=6$; day-FE over 5 days ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 32.54, \quad p = 1.28e-05 \quad *** \quad | \quad \text{BH-FDR } p = 0.000116 \quad ***$$

Direction B – ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..6) d_j \cdot y_{t-j} + \sum(d=1..4) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=6$; day-FE over 5 days ; hac-panel SE (bw=6).

$$H_0: d_1 = \dots = d_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 8.33, \quad p = 0.215 \quad | \quad \text{BH-FDR } p = 0.568$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..6) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=6$; sample spans 5 days (probit: no day-FE) ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 8.92, \quad p = 0.178 \quad | \quad \text{BH-FDR } p = 0.303$$

$\Rightarrow$ event verdict: Pooled Granger-leads ETF.

GRANGER — ELECTION_trump_fav (combined pooled signal) x VCR

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). $x=\Delta$ prob(S), y =ETF log return.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..6) b_j \cdot x_{t-j} + \sum(d=1..21) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=6$; day-FE over 22 days ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 11.86, \quad p = 0.0651 * \quad | \quad \text{BH-FDR } p = 0.172$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..6) d_j \cdot y_{t-j} + \sum(d=1..21) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=6$; day-FE over 22 days ; hac-panel SE (bw=6).

$$H_0: d_1 = \dots = d_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 6.47, \quad p = 0.372 \quad | \quad \text{BH-FDR } p = 0.735$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..6) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=6$; sample spans 22 days (probit: no day-FE) ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 3.48, \quad p = 0.747 \quad | \quad \text{BH-FDR } p = 0.864$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..6) b_j \cdot x_{t-j} + \sum(d=1..4) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=6$; day-FE over 5 days ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 11.08, \quad p = 0.086 * \quad | \quad \text{BH-FDR } p = 0.119$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..6) d_j \cdot y_{t-j} + \sum(d=1..4) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=6$; day-FE over 5 days ; hac-panel SE (bw=6).

$$H_0: d_1 = \dots = d_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 7.81, \quad p = 0.253 \quad | \quad \text{BH-FDR } p = 0.568$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..6) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=6$; sample spans 5 days (probit: no day-FE) ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 10.95, \quad p = 0.09 * \quad | \quad \text{BH-FDR } p = 0.231$$

$\Rightarrow$ event verdict: neither direction significant.

GRANGER — ELECTION_trump_fav (combined pooled signal) x VDC

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). $x=\Delta$ prob(S), $y=\text{ETF log return}$.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..6) b_j \cdot x_{t-j} + \sum(d=1..21) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=6$; day-FE over 22 days ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 16.92, \quad p = 0.00959 \text{ ***} \quad | \quad \text{BH-FDR } p = 0.0443 \text{ **}$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..6) d_j \cdot y_{t-j} + \sum(d=1..21) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=6$; day-FE over 22 days ; hac-panel SE (bw=6).

$$H_0: d_1 = \dots = d_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 7.99, \quad p = 0.239 \quad | \quad \text{BH-FDR } p = 0.735$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..6) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=6$; sample spans 22 days (probit: no day-FE) ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 10.56, \quad p = 0.103 \quad | \quad \text{BH-FDR } p = 0.452$$

$\Rightarrow$ calendar verdict: Pooled Granger-leads ETF.

2. EVENT-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..6) b_j \cdot x_{t-j} + \sum(d=1..4) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=6$; day-FE over 5 days ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 15.87, \quad p = 0.0145 \text{ **} \quad | \quad \text{BH-FDR } p = 0.0261 \text{ **}$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..6) d_j \cdot y_{t-j} + \sum(d=1..4) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=6$; day-FE over 5 days ; hac-panel SE (bw=6).

$$H_0: d_1 = \dots = d_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 8.71, \quad p = 0.191 \quad | \quad \text{BH-FDR } p = 0.568$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..6) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=6$; sample spans 5 days (probit: no day-FE) ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 8.80, \quad p = 0.185 \quad | \quad \text{BH-FDR } p = 0.303$$

$\Rightarrow$ event verdict: Pooled Granger-leads ETF.

GRANGER — ELECTION_trump_fav (combined pooled signal) x VDE

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). $x=\Delta$ prob(S), y =ETF log return.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \sum(j=1..6) b_j \cdot x_{t-j} + \sum(d=1..21) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=2$ (BIC) ; cause block $K=6$; day-FE over 22 days ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 9.83, \quad p = 0.132 \quad | \quad \text{BH-FDR } p = 0.272$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..6) d_j \cdot y_{t-j} + \sum(d=1..21) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=6$; day-FE over 22 days ; hac-panel SE (bw=6).

$$H_0: d_1 = \dots = d_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 8.53, \quad p = 0.202 \quad | \quad \text{BH-FDR } p = 0.735$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..6) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=6$; sample spans 22 days (probit: no day-FE) ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 10.44, \quad p = 0.107 \quad | \quad \text{BH-FDR } p = 0.452$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..6) b_j \cdot x_{t-j} + \sum(d=1..4) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=6$; day-FE over 5 days ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 20.93, \quad p = 0.00189 \text{ ***} \quad | \quad \text{BH-FDR } p = 0.0041 \text{ ***}$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..6) d_j \cdot y_{t-j} + \sum(d=1..4) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=6$; day-FE over 5 days ; hac-panel SE (bw=6).

$$H_0: d_1 = \dots = d_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 14.76, \quad p = 0.0222 \text{ **} \quad | \quad \text{BH-FDR } p = 0.269$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..6) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=6$; sample spans 5 days (probit: no day-FE) ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 9.65, \quad p = 0.14 \quad | \quad \text{BH-FDR } p = 0.28$$

$\Rightarrow$ event verdict: bidirectional (both reject).

GRANGER — ELECTION_trump_fav (combined pooled signal) x VFH

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). $x=\Delta$ prob(S), $y=\text{ETF log return}$.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..6) b_j \cdot x_{t-j} + \sum(d=1..21) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=6$; day-FE over 22 days ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 25.51, \quad p = 0.000274 \quad *** \quad | \quad \text{BH-FDR } p = 0.00307 \quad ***$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..6) d_j \cdot y_{t-j} + \sum(d=1..21) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=6$; day-FE over 22 days ; hac-panel SE (bw=6).

$$H_0: d_1 = \dots = d_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 3.10, \quad p = 0.796 \quad | \quad \text{BH-FDR } p = 0.859$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..6) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=6$; sample spans 22 days (probit: no day-FE) ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 22.06, \quad p = 0.00118 \quad *** \quad | \quad \text{BH-FDR } p = 0.0438 \quad **$$

$\Rightarrow$ calendar verdict: Pooled Granger-leads ETF.

2. EVENT-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..6) b_j \cdot x_{t-j} + \sum(d=1..4) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=6$; day-FE over 5 days ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 30.84, \quad p = 2.72e-05 \quad *** \quad | \quad \text{BH-FDR } p = 0.000163 \quad ***$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..6) d_j \cdot y_{t-j} + \sum(d=1..4) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=6$; day-FE over 5 days ; hac-panel SE (bw=6).

$$H_0: d_1 = \dots = d_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 4.64, \quad p = 0.591 \quad | \quad \text{BH-FDR } p = 0.818$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..6) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=6$; sample spans 5 days (probit: no day-FE) ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 19.65, \quad p = 0.00319 \quad *** \quad | \quad \text{BH-FDR } p = 0.0381 \quad **$$

$\Rightarrow$ event verdict: Pooled Granger-leads ETF.

GRANGER — ELECTION_trump_fav (combined pooled signal) x VGT

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). $x=\Delta$ prob(S), y =ETF log return.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \sum(j=1..6) b_j \cdot x_{t-j} + \sum(d=1..21) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=2$ (BIC) ; cause block $K=6$; day-FE over 22 days ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 3.87, \quad p = 0.694 \quad | \quad \text{BH-FDR } p = 0.755$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..6) d_j \cdot y_{t-j} + \sum(d=1..21) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=6$; day-FE over 22 days ; hac-panel SE (bw=6).

$$H_0: d_1 = \dots = d_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 4.12, \quad p = 0.66 \quad | \quad \text{BH-FDR } p = 0.842$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..6) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=6$; sample spans 22 days (probit: no day-FE) ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 1.93, \quad p = 0.926 \quad | \quad \text{BH-FDR } p = 0.963$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..6) b_j \cdot x_{t-j} + \sum(d=1..4) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=6$; day-FE over 5 days ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 9.66, \quad p = 0.14 \quad | \quad \text{BH-FDR } p = 0.18$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..6) d_j \cdot y_{t-j} + \sum(d=1..4) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=6$; day-FE over 5 days ; hac-panel SE (bw=6).

$$H_0: d_1 = \dots = d_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 3.69, \quad p = 0.718 \quad | \quad \text{BH-FDR } p = 0.861$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..6) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=6$; sample spans 5 days (probit: no day-FE) ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 5.02, \quad p = 0.541 \quad | \quad \text{BH-FDR } p = 0.609$$

$\Rightarrow$ event verdict: neither direction significant.

GRANGER — ELECTION_trump_fav (combined pooled signal) x VIS

=====

S = combined pooled signal (members' sign-aligned Δprob, pooled — NOT a price level). x=Δprob(S), y=ETF log return.

Joint Wald H0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member×day. p<0.05 ⇒ leads.

1. CALENDAR-TIME

Direction A — Pooled → ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..6) b_j \cdot x_{t-j} + \sum(d=1..21) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags p=1 (BIC) ; cause block K=6 ; day-FE over 22 days ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 14.18, \quad p = 0.0277^{**} \quad | \quad \text{BH-FDR } p = 0.0817^*$$

Direction B — ETF → Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..6) d_j \cdot y_{t-j} + \sum(d=1..21) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags p=5 (BIC) ; cause block K=6 ; day-FE over 22 days ; hac-panel SE (bw=6).

$$H_0: d_1 = \dots = d_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 6.69, \quad p = 0.35 \quad | \quad \text{BH-FDR } p = 0.735$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..6) b_j \cdot x_{t-j})$$

controls: ADL self-lags p=1 (BIC) ; cause block K=6 ; sample spans 22 days (probit: no day-FE) ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 8.83, \quad p = 0.183 \quad | \quad \text{BH-FDR } p = 0.452$$

=> calendar verdict: Pooled Granger-leads ETF.

2. EVENT-TIME

Direction A — Pooled → ETF:

$$y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..6) b_j \cdot x_{t-j} + \sum(d=1..4) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags p=0 (BIC) ; cause block K=6 ; day-FE over 5 days ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 34.70, \quad p = 4.92e-06^{***} \quad | \quad \text{BH-FDR } p = 8.85e-05^{***}$$

Direction B — ETF → Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..6) d_j \cdot y_{t-j} + \sum(d=1..4) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags p=5 (BIC) ; cause block K=6 ; day-FE over 5 days ; hac-panel SE (bw=6).

$$H_0: d_1 = \dots = d_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 8.02, \quad p = 0.237 \quad | \quad \text{BH-FDR } p = 0.568$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..6) b_j \cdot x_{t-j})$$

controls: ADL self-lags p=0 (BIC) ; cause block K=6 ; sample spans 5 days (probit: no day-FE) ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 14.74, \quad p = 0.0224^{**} \quad | \quad \text{BH-FDR } p = 0.0806^*$$

=> event verdict: Pooled Granger-leads ETF.

GRANGER — ELECTION_trump_fav (combined pooled signal) x VNQ

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). $x=\Delta$ prob(S), y =ETF log return.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \sum(j=1..6) b_j \cdot x_{t-j} + \sum(d=1..21) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=2$ (BIC) ; cause block $K=6$; day-FE over 22 days ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 7.00, \quad p = 0.321 \quad | \quad \text{BH-FDR } p = 0.475$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..6) d_j \cdot y_{t-j} + \sum(d=1..21) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=6$; day-FE over 22 days ; hac-panel SE (bw=6).

$$H_0: d_1 = \dots = d_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 5.73, \quad p = 0.454 \quad | \quad \text{BH-FDR } p = 0.735$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..6) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=6$; sample spans 22 days (probit: no day-FE) ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 7.42, \quad p = 0.284 \quad | \quad \text{BH-FDR } p = 0.552$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..6) b_j \cdot x_{t-j} + \sum(d=1..4) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=6$; day-FE over 5 days ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 20.93, \quad p = 0.00189 \text{ ***} \quad | \quad \text{BH-FDR } p = 0.0041 \text{ ***}$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..6) d_j \cdot y_{t-j} + \sum(d=1..4) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=6$; day-FE over 5 days ; hac-panel SE (bw=6).

$$H_0: d_1 = \dots = d_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 13.98, \quad p = 0.0299 \text{ **} \quad | \quad \text{BH-FDR } p = 0.269$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..6) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=6$; sample spans 4 days (probit: no day-FE) ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 11.08, \quad p = 0.0859 \text{ *} \quad | \quad \text{BH-FDR } p = 0.231$$

$\Rightarrow$ event verdict: bidirectional (both reject).

GRANGER – ELECTION_trump_fav (combined pooled signal) x VOX

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled – NOT a price level). $x=\Delta$ prob(S), y =ETF log return.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A – Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \sum(j=1..6) b_j \cdot x_{t-j} + \sum(d=1..21) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=2$ (BIC) ; cause block $K=6$; day-FE over 22 days ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 8.64, \quad p = 0.195 \quad | \quad \text{BH-FDR } p = 0.38$$

Direction B – ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..6) d_j \cdot y_{t-j} + \sum(d=1..21) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=6$; day-FE over 22 days ; hac-panel SE (bw=6).

$$H_0: d_1 = \dots = d_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 8.87, \quad p = 0.181 \quad | \quad \text{BH-FDR } p = 0.735$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..6) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=6$; sample spans 22 days (probit: no day-FE) ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 4.01, \quad p = 0.676 \quad | \quad \text{BH-FDR } p = 0.817$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

Direction A – Pooled $\rightarrow$ ETF:

$$y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..6) b_j \cdot x_{t-j} + \sum(d=1..4) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=6$; day-FE over 5 days ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 26.64, \quad p = 0.000169 \quad *** \quad | \quad \text{BH-FDR } p = 0.000761 \quad ***$$

Direction B – ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..6) d_j \cdot y_{t-j} + \sum(d=1..4) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=6$; day-FE over 5 days ; hac-panel SE (bw=6).

$$H_0: d_1 = \dots = d_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 6.79, \quad p = 0.341 \quad | \quad \text{BH-FDR } p = 0.614$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..6) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=6$; sample spans 5 days (probit: no day-FE) ; hac-panel SE (bw=6).

$$H_0: b_1 = \dots = b_6 = 0 \rightarrow \text{Wald } \chi^2(6) = 18.96, \quad p = 0.00423 \quad *** \quad | \quad \text{BH-FDR } p = 0.0381 \quad **$$

$\Rightarrow$ event verdict: Pooled Granger-leads ETF.

MERGE FOMC_easing – Granger overview

MERGE pooled signal – GRANGER overview : FOMC_easing

=====

Each pooled signal = members' sign-aligned Δ prob pooled (member fixed effects, hac-panel SE by member $\times$ day). 'Sign-aligned' means each member's Δ prob is multiplied by ± 1 so that a positive value always means the same real-world direction (e.g. a MIN-side contract is flipped). The CONTINUOUS magnitude of Δ prob is kept – the moves are NOT discretized to ± 1 .

Joint Wald per direction; S=combined pooled signal, E=ETF. $p < 0.05 \Rightarrow$ that side leads.

ETF	cal S→E	cal E→S	evt S→E	evt E→S	verdict(cal / evt)
VCR	0.31	0.22	0.6	0.84	- / -
VDC	0.41	0.25	0.079*	0.33	- / -
VDE	0.11	0.83	0.21	0.66	- / -
VFH	0.54	0.81	0.002***	0.43	- / S→E
VGT	0.79	0.024**	0.69	0.45	E→S / -
VIS	0.8	0.31	0.049**	0.9	- / S→E
VNQ	0.43	0.45	0.0014***	0.1	- / S→E
VOX	0.31	0.35	0.68	0.8	- / -
VPU	0.012**	0.32	0.0012***	0.13	S→E / S→E

Per-ETF pseudo-pairs (combined pooled signal $\times$ each ETF) follow, each with explicit formulas + joint Wald.

GRANGER — FOMC_easing (combined pooled signal) x VCR

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). $x=\Delta$ prob(S), y =ETF log return.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \phi_3 \cdot y_{t-3} + \phi_4 \cdot y_{t-4} + \phi_5 \cdot y_{t-5} + \phi_6 \cdot y_{t-6} + \sum(j=1..10) b_j \cdot x_{t-j} + \sum(d=1..74) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=6$ (BIC) ; cause block $K=10$; day-FE over 75 days ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 11.70, \quad p = 0.306 \quad | \quad \text{BH-FDR } p = 0.475$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..10) d_j \cdot y_{t-j} + \sum(d=1..74) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=10$; day-FE over 75 days ; hac-panel SE (bw=10).

$$H_0: d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 13.12, \quad p = 0.217 \quad | \quad \text{BH-FDR } p = 0.735$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; sample spans 75 days (probit: no day-FE) ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 9.98, \quad p = 0.442 \quad | \quad \text{BH-FDR } p = 0.654$$

=> calendar verdict: neither direction significant.

2. EVENT-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j} + \sum(d=1..42) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; day-FE over 43 days ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 8.28, \quad p = 0.602 \quad | \quad \text{BH-FDR } p = 0.677$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \psi_6 \cdot x_{t-6} + \sum(j=1..10) d_j \cdot y_{t-j} + \sum(d=1..42) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=6$ (BIC) ; cause block $K=10$; day-FE over 43 days ; hac-panel SE (bw=10).

$$H_0: d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 5.70, \quad p = 0.84 \quad | \quad \text{BH-FDR } p = 0.889$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; sample spans 43 days (probit: no day-FE) ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 13.13, \quad p = 0.216 \quad | \quad \text{BH-FDR } p = 0.325$$

=> event verdict: neither direction significant.

GRANGER — FOMC_easing (combined pooled signal) x VDC

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). $x=\Delta\text{prob}(S)$, $y=\text{ETF log return}$.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j} + \sum(d=1..74) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; day-FE over 75 days ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 10.33, \quad p = 0.412 \quad | \quad \text{BH-FDR } p = 0.522$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..10) d_j \cdot y_{t-j} + \sum(d=1..74) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=10$; day-FE over 75 days ; hac-panel SE (bw=10).

$$H_0: d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 12.58, \quad p = 0.248 \quad | \quad \text{BH-FDR } p = 0.735$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; sample spans 75 days (probit: no day-FE) ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 10.80, \quad p = 0.373 \quad | \quad \text{BH-FDR } p = 0.627$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j} + \sum(d=1..42) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; day-FE over 43 days ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 16.81, \quad p = 0.0786 * \quad | \quad \text{BH-FDR } p = 0.118$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \psi_6 \cdot x_{t-6} + \sum(j=1..10) d_j \cdot y_{t-j} + \sum(d=1..42) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=6$ (BIC) ; cause block $K=10$; day-FE over 43 days ; hac-panel SE (bw=10).

$$H_0: d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 11.35, \quad p = 0.331 \quad | \quad \text{BH-FDR } p = 0.614$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; sample spans 41 days (probit: no day-FE) ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 10.96, \quad p = 0.36 \quad | \quad \text{BH-FDR } p = 0.499$$

$\Rightarrow$ event verdict: neither direction significant.

GRANGER — FOMC_easing (combined pooled signal) x VDE

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). $x=\Delta$ prob(S), y =ETF log return.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j} + \sum(d=1..74) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; day-FE over 75 days ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 15.56, \quad p = 0.113 \quad | \quad \text{BH-FDR } p = 0.246$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..10) d_j \cdot y_{t-j} + \sum(d=1..74) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=10$; day-FE over 75 days ; hac-panel SE (bw=10).

$$H_0: d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 5.76, \quad p = 0.835 \quad | \quad \text{BH-FDR } p = 0.859$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; sample spans 75 days (probit: no day-FE) ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 7.43, \quad p = 0.685 \quad | \quad \text{BH-FDR } p = 0.817$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..10) b_j \cdot x_{t-j} + \sum(d=1..42) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=10$; day-FE over 43 days ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 13.22, \quad p = 0.212 \quad | \quad \text{BH-FDR } p = 0.254$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \psi_6 \cdot x_{t-6} + \sum(j=1..10) d_j \cdot y_{t-j} + \sum(d=1..42) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=6$ (BIC) ; cause block $K=10$; day-FE over 43 days ; hac-panel SE (bw=10).

$$H_0: d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 7.73, \quad p = 0.655 \quad | \quad \text{BH-FDR } p = 0.843$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..10) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=10$; sample spans 41 days (probit: no day-FE) ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 9.27, \quad p = 0.507 \quad | \quad \text{BH-FDR } p = 0.608$$

$\Rightarrow$ event verdict: neither direction significant.

GRANGER — FOMC_easing (combined pooled signal) x VFH

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). $x=\Delta$ prob(S), y =ETF log return.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \phi_3 \cdot y_{t-3} + \phi_4 \cdot y_{t-4} + \phi_5 \cdot y_{t-5} + \sum(j=1..10) b_j \cdot x_{t-j} + \sum(d=1..74) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=10$; day-FE over 75 days ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 8.91, \quad p = 0.54 \quad | \quad \text{BH-FDR } p = 0.606$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..10) d_j \cdot y_{t-j} + \sum(d=1..74) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=10$; day-FE over 75 days ; hac-panel SE (bw=10).

$$H_0: d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 6.02, \quad p = 0.813 \quad | \quad \text{BH-FDR } p = 0.859$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; sample spans 75 days (probit: no day-FE) ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 5.91, \quad p = 0.823 \quad | \quad \text{BH-FDR } p = 0.905$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j} + \sum(d=1..42) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; day-FE over 43 days ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 27.66, \quad p = 0.00205 \quad *** \quad | \quad \text{BH-FDR } p = 0.0041 \quad ***$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \psi_6 \cdot x_{t-6} + \sum(j=1..10) d_j \cdot y_{t-j} + \sum(d=1..42) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=6$ (BIC) ; cause block $K=10$; day-FE over 43 days ; hac-panel SE (bw=10).

$$H_0: d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 10.13, \quad p = 0.429 \quad | \quad \text{BH-FDR } p = 0.675$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; sample spans 42 days (probit: no day-FE) ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 20.99, \quad p = 0.0211 \quad ** \quad | \quad \text{BH-FDR } p = 0.0806 \quad *$$

$\Rightarrow$ event verdict: Pooled Granger-leads ETF.

GRANGER — FOMC_easing (combined pooled signal) x VGT

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). $x=\Delta$ prob(S), y =ETF log return.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \phi_3 \cdot y_{t-3} + \phi_4 \cdot y_{t-4} + \phi_5 \cdot y_{t-5} + \phi_6 \cdot y_{t-6} + \sum(j=1..10) b_j \cdot x_{t-j} + \sum(d=1..74) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=6$ (BIC) ; cause block $K=10$; day-FE over 75 days ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 6.24, \quad p = 0.795 \quad | \quad \text{BH-FDR } p = 0.819$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..10) d_j \cdot y_{t-j} + \sum(d=1..74) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=10$; day-FE over 75 days ; hac-panel SE (bw=10).

$$H_0: d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 20.60, \quad p = 0.024^{**} \quad | \quad \text{BH-FDR } p = 0.735$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \sum(j=1..10) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=2$ (BIC) ; cause block $K=10$; sample spans 75 days (probit: no day-FE) ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 3.25, \quad p = 0.975 \quad | \quad \text{BH-FDR } p = 0.975$$

$\Rightarrow$ calendar verdict: ETF Granger-leads Pooled.

2. EVENT-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j} + \sum(d=1..42) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; day-FE over 43 days ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 7.38, \quad p = 0.689 \quad | \quad \text{BH-FDR } p = 0.689$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \psi_6 \cdot x_{t-6} + \sum(j=1..10) d_j \cdot y_{t-j} + \sum(d=1..42) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=6$ (BIC) ; cause block $K=10$; day-FE over 43 days ; hac-panel SE (bw=10).

$$H_0: d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 9.89, \quad p = 0.45 \quad | \quad \text{BH-FDR } p = 0.675$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..10) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=10$; sample spans 43 days (probit: no day-FE) ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 10.39, \quad p = 0.407 \quad | \quad \text{BH-FDR } p = 0.523$$

$\Rightarrow$ event verdict: neither direction significant.

GRANGER — FOMC_easing (combined pooled signal) x VIS

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). $x=\Delta$ prob(S), y =ETF log return.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \phi_3 \cdot y_{t-3} + \phi_4 \cdot y_{t-4} + \phi_5 \cdot y_{t-5} + \phi_6 \cdot y_{t-6} + \sum_{j=1..10} b_j \cdot x_{t-j} + \sum_{d=1..74} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=6$ (BIC) ; cause block $K=10$; day-FE over 75 days ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 6.21, \quad p = 0.797 \quad | \quad \text{BH-FDR } p = 0.819$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum_{j=1..10} d_j \cdot y_{t-j} + \sum_{d=1..74} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=10$; day-FE over 75 days ; hac-panel SE (bw=10).

$$H_0: d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 11.59, \quad p = 0.313 \quad | \quad \text{BH-FDR } p = 0.735$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..10} b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; sample spans 75 days (probit: no day-FE) ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 8.69, \quad p = 0.561 \quad | \quad \text{BH-FDR } p = 0.742$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..10} b_j \cdot x_{t-j} + \sum_{d=1..42} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; day-FE over 43 days ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 18.37, \quad p = 0.049^{**} \quad | \quad \text{BH-FDR } p = 0.0802^*$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \psi_6 \cdot x_{t-6} + \sum_{j=1..10} d_j \cdot y_{t-j} + \sum_{d=1..42} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=6$ (BIC) ; cause block $K=10$; day-FE over 43 days ; hac-panel SE (bw=10).

$$H_0: d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 4.79, \quad p = 0.905 \quad | \quad \text{BH-FDR } p = 0.905$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum_{j=1..10} b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=10$; sample spans 42 days (probit: no day-FE) ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 8.21, \quad p = 0.608 \quad | \quad \text{BH-FDR } p = 0.644$$

$\Rightarrow$ event verdict: Pooled Granger-leads ETF.

GRANGER — FOMC_easing (combined pooled signal) x VNQ

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). $x=\Delta$ prob(S), y =ETF log return.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \phi_3 \cdot y_{t-3} + \phi_4 \cdot y_{t-4} + \phi_5 \cdot y_{t-5} + \phi_6 \cdot y_{t-6} + \sum(j=1..10) b_j \cdot x_{t-j} + \sum(d=1..74) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=6$ (BIC) ; cause block $K=10$; day-FE over 75 days ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 10.16, \quad p = 0.427 \quad | \quad \text{BH-FDR } p = 0.522$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..10) d_j \cdot y_{t-j} + \sum(d=1..74) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=10$; day-FE over 75 days ; hac-panel SE (bw=10).

$$H_0: d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 9.95, \quad p = 0.445 \quad | \quad \text{BH-FDR } p = 0.735$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; sample spans 75 days (probit: no day-FE) ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 14.15, \quad p = 0.166 \quad | \quad \text{BH-FDR } p = 0.452$$

=> calendar verdict: neither direction significant.

2. EVENT-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j} + \sum(d=1..42) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; day-FE over 43 days ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 28.76, \quad p = 0.00136 \quad *** \quad | \quad \text{BH-FDR } p = 0.00408 \quad ***$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \psi_6 \cdot x_{t-6} + \sum(j=1..10) d_j \cdot y_{t-j} + \sum(d=1..42) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=6$ (BIC) ; cause block $K=10$; day-FE over 43 days ; hac-panel SE (bw=10).

$$H_0: d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 15.85, \quad p = 0.104 \quad | \quad \text{BH-FDR } p = 0.568$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; sample spans 42 days (probit: no day-FE) ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 24.25, \quad p = 0.00697 \quad *** \quad | \quad \text{BH-FDR } p = 0.0418 \quad **$$

=> event verdict: Pooled Granger-leads ETF.

GRANGER — FOMC_easing (combined pooled signal) x VOX

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). $x=\Delta$ prob(S), y =ETF log return.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \phi_3 \cdot y_{t-3} + \phi_4 \cdot y_{t-4} + \phi_5 \cdot y_{t-5} + \sum(j=1..10) b_j \cdot x_{t-j} + \sum(d=1..74) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=10$; day-FE over 75 days ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 11.65, \quad p = 0.309 \quad | \quad \text{BH-FDR } p = 0.475$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..10) d_j \cdot y_{t-j} + \sum(d=1..74) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=10$; day-FE over 75 days ; hac-panel SE (bw=10).

$$H_0: d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 11.08, \quad p = 0.352 \quad | \quad \text{BH-FDR } p = 0.735$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; sample spans 75 days (probit: no day-FE) ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 14.05, \quad p = 0.171 \quad | \quad \text{BH-FDR } p = 0.452$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j} + \sum(d=1..42) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; day-FE over 43 days ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 7.51, \quad p = 0.677 \quad | \quad \text{BH-FDR } p = 0.689$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \psi_6 \cdot x_{t-6} + \sum(j=1..10) d_j \cdot y_{t-j} + \sum(d=1..42) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=6$ (BIC) ; cause block $K=10$; day-FE over 43 days ; hac-panel SE (bw=10).

$$H_0: d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 6.21, \quad p = 0.797 \quad | \quad \text{BH-FDR } p = 0.889$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; sample spans 42 days (probit: no day-FE) ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 4.19, \quad p = 0.938 \quad | \quad \text{BH-FDR } p = 0.938$$

$\Rightarrow$ event verdict: neither direction significant.

GRANGER — FOMC_easing (combined pooled signal) x VPU

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). $x=\Delta$ prob(S), $y=\text{ETF log return}$.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j} + \sum(d=1..74) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; day-FE over 75 days ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 22.67, \quad p = 0.012^{**} \quad | \quad \text{BH-FDR } p = 0.0495^{**}$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \sum(j=1..10) d_j \cdot y_{t-j} + \sum(d=1..74) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=5$ (BIC) ; cause block $K=10$; day-FE over 75 days ; hac-panel SE (bw=10).

$$H_0: d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 11.50, \quad p = 0.32 \quad | \quad \text{BH-FDR } p = 0.735$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..10) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=10$; sample spans 75 days (probit: no day-FE) ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 21.26, \quad p = 0.0194^{**} \quad | \quad \text{BH-FDR } p = 0.154$$

$\Rightarrow$ calendar verdict: Pooled Granger-leads ETF.

2. EVENT-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..10) b_j \cdot x_{t-j} + \sum(d=1..42) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=10$; day-FE over 43 days ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 29.20, \quad p = 0.00116^{***} \quad | \quad \text{BH-FDR } p = 0.00408^{***}$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \psi_2 \cdot x_{t-2} + \psi_3 \cdot x_{t-3} + \psi_4 \cdot x_{t-4} + \psi_5 \cdot x_{t-5} + \psi_6 \cdot x_{t-6} + \sum(j=1..10) d_j \cdot y_{t-j} + \sum(d=1..42) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=6$ (BIC) ; cause block $K=10$; day-FE over 43 days ; hac-panel SE (bw=10).

$$H_0: d_1 = \dots = d_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 14.94, \quad p = 0.134 \quad | \quad \text{BH-FDR } p = 0.568$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..10) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=10$; sample spans 43 days (probit: no day-FE) ; hac-panel SE (bw=10).

$$H_0: b_1 = \dots = b_{10} = 0 \rightarrow \text{Wald } \chi^2(10) = 15.28, \quad p = 0.122 \quad | \quad \text{BH-FDR } p = 0.275$$

$\Rightarrow$ event verdict: Pooled Granger-leads ETF.

MERGE pooled signal – GRANGER overview : GAS_above

MERGE GAS_above – Granger overview

=====

Each pooled signal = members' sign-aligned Δ prob pooled (member fixed effects, hac-panel SE by member×day). 'Sign-aligned' means each member's Δ prob is multiplied by ± 1 so that a positive value always means the same real-world direction (e.g. a MIN-side contract is flipped). The CONTINUOUS magnitude of Δ prob is kept – the moves are NOT discretized to ± 1 .

Joint Wald per direction; S=combined pooled signal, E=ETF. $p < 0.05 \Rightarrow$ that side leads.

ETF	cal S→E	cal E→S	evt S→E	evt E→S	verdict(cal / evt)
VAW	0.0017***	0.61	-	-	S→E / -
VCR	2.8e-08***	0.4	-	-	S→E / -
VDC	0.029**	0.49	-	-	S→E / -
VDE	0.36	0.75	-	-	- / -
VFH	0.44	0.78	-	-	- / -
VGT	0.0018***	0.46	-	-	S→E / -
VHT	0.086*	0.23	-	-	- / -
VIS	0.016**	0.84	-	-	S→E / -
VNQ	0.0037***	0.15	-	-	S→E / -
VOX	0.00033***	0.5	-	-	S→E / -
VPU	0.0003***	0.48	-	-	S→E / -

Per-ETF pseudo-pairs (combined pooled signal × each ETF) follow, each with explicit formulas + joint Wald.

GRANGER — GAS_above (combined pooled signal) x VAW

=====

S = combined pooled signal (members' sign-aligned Δprob , pooled — NOT a price level). $x=\Delta\text{prob}(S)$, $y=\text{ETF log return}$.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..4) b_j \cdot x_{t-j} + \sum(d=1..24) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$H_0: b_1 = \dots = b_4 = 0 \rightarrow$ Wald $\chi^2(4) = 17.26$, $p = 0.00172$ *** | BH-FDR $p = 0.0111$ **

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum(j=1..4) d_j \cdot y_{t-j} + \sum(d=1..24) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$H_0: d_1 = \dots = d_4 = 0 \rightarrow$ Wald $\chi^2(4) = 2.67$, $p = 0.615$ | BH-FDR $p = 0.812$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..4) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=4$; sample spans 25 days (probit: no day-FE) ; hac-panel SE (bw=4).

$H_0: b_1 = \dots = b_4 = 0 \rightarrow$ Wald $\chi^2(4) = 7.40$, $p = 0.116$ | BH-FDR $p = 0.452$

$\Rightarrow$ calendar verdict: Pooled Granger-leads ETF.

2. EVENT-TIME

(not estimable)

GRANGER — GAS_above (combined pooled signal) x VCR

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). x= Δ prob(S), y=ETF log return.

Joint Wald H0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member×day. p<0.05 ⇒ leads.

1. CALENDAR-TIME

Direction A — Pooled → ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j} + \sum_{d=1..24} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags p=1 (BIC) ; cause block K=4 ; day-FE over 25 days ; hac-panel SE (bw=4).

H0: $b_1 = \dots = b_4 = 0$ → Wald $\chi^2(4) = 40.88$, p = 2.85e-08 *** | BH-FDR p = 1.05e-06 ***

Direction B — ETF → Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum_{j=1..4} d_j \cdot y_{t-j} + \sum_{d=1..24} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags p=1 (BIC) ; cause block K=4 ; day-FE over 25 days ; hac-panel SE (bw=4).

H0: $d_1 = \dots = d_4 = 0$ → Wald $\chi^2(4) = 4.06$, p = 0.397 | BH-FDR p = 0.735

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j})$$

controls: ADL self-lags p=1 (BIC) ; cause block K=4 ; sample spans 25 days (probit: no day-FE) ; hac-panel SE (bw=4).

H0: $b_1 = \dots = b_4 = 0$ → Wald $\chi^2(4) = 4.67$, p = 0.322 | BH-FDR p = 0.568

=> calendar verdict: Pooled Granger-leads ETF.

2. EVENT-TIME

(not estimable)

GRANGER — GAS_above (combined pooled signal) x VDC

=====

S = combined pooled signal (members' sign-aligned Δprob , pooled — NOT a price level). $x=\Delta\text{prob}(S)$, $y=\text{ETF log return}$.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \sum_{j=1..4} b_j \cdot x_{t-j} + \sum_{d=1..24} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=2$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$H_0: b_1 = \dots = b_4 = 0 \rightarrow$ Wald $\chi^2(4) = 10.82$, $p = 0.0287^{**}$ | BH-FDR $p = 0.0817^*$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum_{j=1..4} d_j \cdot y_{t-j} + \sum_{d=1..24} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$H_0: d_1 = \dots = d_4 = 0 \rightarrow$ Wald $\chi^2(4) = 3.45$, $p = 0.486$ | BH-FDR $p = 0.735$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; sample spans 24 days (probit: no day-FE) ; hac-panel SE (bw=4).

$H_0: b_1 = \dots = b_4 = 0 \rightarrow$ Wald $\chi^2(4) = 4.11$, $p = 0.391$ | BH-FDR $p = 0.629$

$\Rightarrow$ calendar verdict: Pooled Granger-leads ETF.

2. EVENT-TIME

(not estimable)

GRANGER — GAS_above (combined pooled signal) x VDE

=====

S = combined pooled signal (members' sign-aligned Δprob , pooled — NOT a price level). $x=\Delta\text{prob}(S)$, $y=\text{ETF log return}$.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j} + \sum_{d=1..24} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$$H_0: b_1 = \dots = b_4 = 0 \quad \rightarrow \quad \text{Wald } \chi^2(4) = 4.33, \quad p = 0.363 \quad | \quad \text{BH-FDR } p = 0.498$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum_{j=1..4} d_j \cdot y_{t-j} + \sum_{d=1..24} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$$H_0: d_1 = \dots = d_4 = 0 \quad \rightarrow \quad \text{Wald } \chi^2(4) = 1.95, \quad p = 0.745 \quad | \quad \text{BH-FDR } p = 0.859$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; sample spans 25 days (probit: no day-FE) ; hac-panel SE (bw=4).

$$H_0: b_1 = \dots = b_4 = 0 \quad \rightarrow \quad \text{Wald } \chi^2(4) = 5.88, \quad p = 0.208 \quad | \quad \text{BH-FDR } p = 0.476$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

(not estimable)

GRANGER — GAS_above (combined pooled signal) x VFH

=====

S = combined pooled signal (members' sign-aligned Δprob , pooled — NOT a price level). $x=\Delta\text{prob}(S)$, $y=\text{ETF log return}$.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j} + \sum_{d=1..24} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$$H_0: b_1 = \dots = b_4 = 0 \quad \rightarrow \quad \text{Wald } \chi^2(4) = 3.78, \quad p = 0.437 \quad | \quad \text{BH-FDR } p = 0.522$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum_{j=1..4} d_j \cdot y_{t-j} + \sum_{d=1..24} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$$H_0: d_1 = \dots = d_4 = 0 \quad \rightarrow \quad \text{Wald } \chi^2(4) = 1.75, \quad p = 0.781 \quad | \quad \text{BH-FDR } p = 0.859$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; sample spans 25 days (probit: no day-FE) ; hac-panel SE (bw=4).

$$H_0: b_1 = \dots = b_4 = 0 \quad \rightarrow \quad \text{Wald } \chi^2(4) = 3.49, \quad p = 0.48 \quad | \quad \text{BH-FDR } p = 0.683$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

(not estimable)

GRANGER — GAS_above (combined pooled signal) x VGT

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). x= Δ prob(S), y=ETF log return.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p < 0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j} + \sum_{d=1..24} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$H_0: b_1 = \dots = b_4 = 0 \rightarrow$ Wald $\chi^2(4) = 17.16$, $p = 0.0018$ *** | BH-FDR $p = 0.0111$ **

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum_{j=1..4} d_j \cdot y_{t-j} + \sum_{d=1..24} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$H_0: d_1 = \dots = d_4 = 0 \rightarrow$ Wald $\chi^2(4) = 3.62$, $p = 0.46$ | BH-FDR $p = 0.735$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum_{j=1..4} b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=4$; sample spans 25 days (probit: no day-FE) ; hac-panel SE (bw=4).

$H_0: b_1 = \dots = b_4 = 0 \rightarrow$ Wald $\chi^2(4) = 11.57$, $p = 0.0208$ ** | BH-FDR $p = 0.154$

$\Rightarrow$ calendar verdict: Pooled Granger-leads ETF.

2. EVENT-TIME

(not estimable)

GRANGER — GAS_above (combined pooled signal) x VHT

=====

S = combined pooled signal (members' sign-aligned Δprob , pooled — NOT a price level). $x=\Delta\text{prob}(S)$, $y=\text{ETF log return}$.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j} + \sum_{d=1..24} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$$H_0: b_1 = \dots = b_4 = 0 \quad \rightarrow \quad \text{Wald } \chi^2(4) = 8.15, \quad p = 0.0863 * \quad | \quad \text{BH-FDR } p = 0.2$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum_{j=1..4} d_j \cdot y_{t-j} + \sum_{d=1..24} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$$H_0: d_1 = \dots = d_4 = 0 \quad \rightarrow \quad \text{Wald } \chi^2(4) = 5.59, \quad p = 0.232 \quad | \quad \text{BH-FDR } p = 0.735$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; sample spans 25 days (probit: no day-FE) ; hac-panel SE (bw=4).

$$H_0: b_1 = \dots = b_4 = 0 \quad \rightarrow \quad \text{Wald } \chi^2(4) = 15.28, \quad p = 0.00415 *** \quad | \quad \text{BH-FDR } p = 0.0559 *$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

(not estimable)

GRANGER — GAS_above (combined pooled signal) x VIS

=====

S = combined pooled signal (members' sign-aligned Δprob , pooled — NOT a price level). $x=\Delta\text{prob}(S)$, $y=\text{ETF log return}$.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \phi_2 \cdot y_{t-2} + \sum_{j=1..4} b_j \cdot x_{t-j} + \sum_{d=1..24} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=2$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$H_0: b_1 = \dots = b_4 = 0 \rightarrow$ Wald $\chi^2(4) = 12.17$, $p = 0.0161$ ** | BH-FDR $p = 0.0542$ *

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum_{j=1..4} d_j \cdot y_{t-j} + \sum_{d=1..24} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$H_0: d_1 = \dots = d_4 = 0 \rightarrow$ Wald $\chi^2(4) = 1.45$, $p = 0.836$ | BH-FDR $p = 0.859$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; sample spans 25 days (probit: no day-FE) ; hac-panel SE (bw=4).

$H_0: b_1 = \dots = b_4 = 0 \rightarrow$ Wald $\chi^2(4) = 10.36$, $p = 0.0347$ ** | BH-FDR $p = 0.214$

$\Rightarrow$ calendar verdict: Pooled Granger-leads ETF.

2. EVENT-TIME

(not estimable)

GRANGER — GAS_above (combined pooled signal) x VNQ

=====

S = combined pooled signal (members' sign-aligned Δprob , pooled — NOT a price level). $x=\Delta\text{prob}(S)$, $y=\text{ETF log return}$.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum(j=1..4) b_j \cdot x_{t-j} + \sum(d=1..24) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$H_0: b_1 = \dots = b_4 = 0 \rightarrow$ Wald $\chi^2(4) = 15.56$, $p = 0.00366$ *** | BH-FDR $p = 0.0194$ **

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum(j=1..4) d_j \cdot y_{t-j} + \sum(d=1..24) \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$H_0: d_1 = \dots = d_4 = 0 \rightarrow$ Wald $\chi^2(4) = 6.79$, $p = 0.147$ | BH-FDR $p = 0.735$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum(j=1..4) b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=4$; sample spans 25 days (probit: no day-FE) ; hac-panel SE (bw=4).

$H_0: b_1 = \dots = b_4 = 0 \rightarrow$ Wald $\chi^2(4) = 3.85$, $p = 0.426$ | BH-FDR $p = 0.654$

$\Rightarrow$ calendar verdict: Pooled Granger-leads ETF.

2. EVENT-TIME

(not estimable)

GRANGER — GAS_above (combined pooled signal) x VOX

=====

S = combined pooled signal (members' sign-aligned Δprob , pooled — NOT a price level). $x=\Delta\text{prob}(S)$, $y=\text{ETF log return}$.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j} + \sum_{d=1..24} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$H_0: b_1 = \dots = b_4 = 0 \rightarrow$ Wald $\chi^2(4) = 20.90$, $p = 0.000332$ *** | BH-FDR $p = 0.00307$ ***

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum_{j=1..4} d_j \cdot y_{t-j} + \sum_{d=1..24} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$H_0: d_1 = \dots = d_4 = 0 \rightarrow$ Wald $\chi^2(4) = 3.38$, $p = 0.496$ | BH-FDR $p = 0.735$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum_{j=1..4} b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=4$; sample spans 25 days (probit: no day-FE) ; hac-panel SE (bw=4).

$H_0: b_1 = \dots = b_4 = 0 \rightarrow$ Wald $\chi^2(4) = 5.27$, $p = 0.261$ | BH-FDR $p = 0.536$

$\Rightarrow$ calendar verdict: Pooled Granger-leads ETF.

2. EVENT-TIME

(not estimable)

GRANGER — GAS_above (combined pooled signal) x VPU

=====

S = combined pooled signal (members' sign-aligned Δprob , pooled — NOT a price level). $x=\Delta\text{prob}(S)$, $y=\text{ETF log return}$.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j} + \sum_{d=1..24} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$H_0: b_1 = \dots = b_4 = 0 \rightarrow$ Wald $\chi^2(4) = 21.09$, $p = 0.000303$ *** | BH-FDR $p = 0.00307$ ***

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + \psi_1 \cdot x_{t-1} + \sum_{j=1..4} d_j \cdot y_{t-j} + \sum_{d=1..24} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 25 days ; hac-panel SE (bw=4).

$H_0: d_1 = \dots = d_4 = 0 \rightarrow$ Wald $\chi^2(4) = 3.50$, $p = 0.477$ | BH-FDR $p = 0.735$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; sample spans 25 days (probit: no day-FE) ; hac-panel SE (bw=4).

$H_0: b_1 = \dots = b_4 = 0 \rightarrow$ Wald $\chi^2(4) = 3.36$, $p = 0.5$ | BH-FDR $p = 0.685$

$\Rightarrow$ calendar verdict: Pooled Granger-leads ETF.

2. EVENT-TIME

(not estimable)

MERGE pooled signal – GRANGER overview : APPROVAL_strength – Granger overview

=====

Each pooled signal = members' sign-aligned Δprob pooled (member fixed effects, hac-panel SE by member×day). 'Sign-aligned' means each member's Δprob is multiplied by ±1 so that a positive value always means the same real-world direction (e.g. a MIN-side contract is flipped). The CONTINUOUS magnitude of Δprob is kept – the moves are NOT discretized to ±1.

Joint Wald per direction; S=combined pooled signal, E=ETF. p<0.05 ⇒ that side leads.

ETF	cal S→E	cal E→S	evt S→E	evt E→S	verdict(cal / evt)
VAW	0.29	0.72	-	-	- / -
VCR	0.23	0.32	-	-	- / -
VFH	0.93	0.26	-	-	- / -
VGT	0.081*	0.33	-	-	- / -
VHT	0.31	0.6	-	-	- / -
VIS	0.35	0.43	-	-	- / -
VNQ	0.38	0.92	-	-	- / -
VOX	0.015**	0.61	-	-	S→E / -

Per-ETF pseudo-pairs (combined pooled signal × each ETF) follow, each with explicit formulas + joint Wald.

GRANGER — APPROVAL_strength (combined pooled signal) x VAW

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). $x=\Delta$ prob(S), $y=\text{ETF log return}$.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j} + \sum_{d=1..30} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 31 days ; hac-panel SE (bw=4).

$$H_0: b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 4.99, \quad p = 0.288 \quad | \quad \text{BH-FDR } p = 0.475$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + (\text{no } x \text{ self-lag term: BIC chose } p=0) + \sum_{j=1..4} d_j \cdot y_{t-j} + \sum_{d=1..30} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=4$; day-FE over 31 days ; hac-panel SE (bw=4).

$$H_0: d_1 = \dots = d_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 2.07, \quad p = 0.723 \quad | \quad \text{BH-FDR } p = 0.859$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum_{j=1..4} b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=4$; sample spans 31 days (probit: no day-FE) ; hac-panel SE (bw=4).

$$H_0: b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 6.37, \quad p = 0.173 \quad | \quad \text{BH-FDR } p = 0.452$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

(not estimable)

GRANGER — APPROVAL_strength (combined pooled signal) x VCR

=====

S = combined pooled signal (members' sign-aligned Δprob , pooled — NOT a price level). $x=\Delta\text{prob}(S)$, $y=\text{ETF log return}$.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j} + \sum_{d=1..30} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 31 days ; hac-panel SE (bw=4).

$$H_0: b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 5.63, \quad p = 0.228 \quad | \quad \text{BH-FDR } p = 0.422$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + (\text{no } x \text{ self-lag term: BIC chose } p=0) + \sum_{j=1..4} d_j \cdot y_{t-j} + \sum_{d=1..30} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=4$; day-FE over 31 days ; hac-panel SE (bw=4).

$$H_0: d_1 = \dots = d_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 4.70, \quad p = 0.32 \quad | \quad \text{BH-FDR } p = 0.735$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum_{j=1..4} b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=4$; sample spans 31 days (probit: no day-FE) ; hac-panel SE (bw=4).

$$H_0: b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 1.47, \quad p = 0.832 \quad | \quad \text{BH-FDR } p = 0.905$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

(not estimable)

GRANGER — APPROVAL_strength (combined pooled signal) x VFH

=====

S = combined pooled signal (members' sign-aligned Δprob , pooled — NOT a price level). $x=\Delta\text{prob}(S)$, $y=\text{ETF log return}$.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j} + \sum_{d=1..30} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 31 days ; hac-panel SE (bw=4).

$$H_0: b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 0.84, \quad p = 0.934 \quad | \quad \text{BH-FDR } p = 0.934$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + (\text{no } x \text{ self-lag term: BIC chose } p=0) + \sum_{j=1..4} d_j \cdot y_{t-j} + \sum_{d=1..30} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=4$; day-FE over 31 days ; hac-panel SE (bw=4).

$$H_0: d_1 = \dots = d_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 5.29, \quad p = 0.259 \quad | \quad \text{BH-FDR } p = 0.735$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; sample spans 31 days (probit: no day-FE) ; hac-panel SE (bw=4).

$$H_0: b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 0.81, \quad p = 0.937 \quad | \quad \text{BH-FDR } p = 0.963$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

(not estimable)

GRANGER — APPROVAL_strength (combined pooled signal) x VGT

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). $x=\Delta$ prob(S), $y=ETF$ log return.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j} + \sum_{d=1..30} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 31 days ; hac-panel SE (bw=4).

$$H_0: b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 8.32, \quad p = 0.0806 * \quad | \quad \text{BH-FDR } p = 0.199$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + (\text{no } x \text{ self-lag term: BIC chose } p=0) + \sum_{j=1..4} d_j \cdot y_{t-j} + \sum_{d=1..30} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=4$; day-FE over 31 days ; hac-panel SE (bw=4).

$$H_0: d_1 = \dots = d_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 4.64, \quad p = 0.327 \quad | \quad \text{BH-FDR } p = 0.735$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum_{j=1..4} b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=4$; sample spans 31 days (probit: no day-FE) ; hac-panel SE (bw=4).

$$H_0: b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 7.11, \quad p = 0.13 \quad | \quad \text{BH-FDR } p = 0.452$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

(not estimable)

GRANGER — APPROVAL_strength (combined pooled signal) x VHT

=====

S = combined pooled signal (members' sign-aligned Δprob , pooled — NOT a price level). $x=\Delta\text{prob}(S)$, $y=\text{ETF log return}$.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j} + \sum_{d=1..30} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 31 days ; hac-panel SE (bw=4).

$$H_0: b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 4.76, \quad p = 0.313 \quad | \quad \text{BH-FDR } p = 0.475$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + (\text{no } x \text{ self-lag term: BIC chose } p=0) + \sum_{j=1..4} d_j \cdot y_{t-j} + \sum_{d=1..30} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=4$; day-FE over 31 days ; hac-panel SE (bw=4).

$$H_0: d_1 = \dots = d_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 2.77, \quad p = 0.597 \quad | \quad \text{BH-FDR } p = 0.812$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + (\text{no } y \text{ self-lag term: BIC chose } p=0) + \sum_{j=1..4} b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=4$; sample spans 31 days (probit: no day-FE) ; hac-panel SE (bw=4).

$$H_0: b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 6.55, \quad p = 0.162 \quad | \quad \text{BH-FDR } p = 0.452$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

(not estimable)

GRANGER — APPROVAL_strength (combined pooled signal) x VIS

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). x= Δ prob(S), y=ETF log return.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p < 0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$y_t = \alpha +$ (no y self-lag term: BIC chose $p=0$) $+ \sum(j=1..4) b_j \cdot x_{t-j} + \sum(d=1..30) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=4$; day-FE over 31 days ; hac-panel SE (bw=4).

$H_0: b_1 = \dots = b_4 = 0 \rightarrow$ Wald $\chi^2(4) = 4.43, \quad p = 0.351 \quad | \quad \text{BH-FDR } p = 0.498$

Direction B — ETF $\rightarrow$ Pooled:

$x_t = \alpha +$ (no x self-lag term: BIC chose $p=0$) $+ \sum(j=1..4) d_j \cdot y_{t-j} + \sum(d=1..30) \gamma_d \cdot \text{Day}_d$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=4$; day-FE over 31 days ; hac-panel SE (bw=4).

$H_0: d_1 = \dots = d_4 = 0 \rightarrow$ Wald $\chi^2(4) = 3.81, \quad p = 0.432 \quad | \quad \text{BH-FDR } p = 0.735$

Direction A (probit, ETF up/down):

$\Pr(y_t > 0) = \Phi(\alpha +$ (no y self-lag term: BIC chose $p=0$) $+ \sum(j=1..4) b_j \cdot x_{t-j})$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=4$; sample spans 31 days (probit: no day-FE) ; hac-panel SE (bw=4).

$H_0: b_1 = \dots = b_4 = 0 \rightarrow$ Wald $\chi^2(4) = 2.57, \quad p = 0.633 \quad | \quad \text{BH-FDR } p = 0.807$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

(not estimable)

GRANGER — APPROVAL_strength (combined pooled signal) x VNQ

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). $x=\Delta$ prob(S), $y=\text{ETF log return}$.

Joint Wald H_0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. $p<0.05 \Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j} + \sum_{d=1..30} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; day-FE over 31 days ; hac-panel SE (bw=4).

$$H_0: b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 4.22, \quad p = 0.377 \quad | \quad \text{BH-FDR } p = 0.498$$

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + (\text{no } x \text{ self-lag term: BIC chose } p=0) + \sum_{j=1..4} d_j \cdot y_{t-j} + \sum_{d=1..30} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags $p=0$ (BIC) ; cause block $K=4$; day-FE over 31 days ; hac-panel SE (bw=4).

$$H_0: d_1 = \dots = d_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 0.93, \quad p = 0.92 \quad | \quad \text{BH-FDR } p = 0.92$$

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j})$$

controls: ADL self-lags $p=1$ (BIC) ; cause block $K=4$; sample spans 31 days (probit: no day-FE) ; hac-panel SE (bw=4).

$$H_0: b_1 = \dots = b_4 = 0 \rightarrow \text{Wald } \chi^2(4) = 4.79, \quad p = 0.309 \quad | \quad \text{BH-FDR } p = 0.568$$

$\Rightarrow$ calendar verdict: neither direction significant.

2. EVENT-TIME

(not estimable)

GRANGER — APPROVAL_strength (combined pooled signal) x VOX

=====

S = combined pooled signal (members' sign-aligned Δ prob, pooled — NOT a price level). x= Δ prob(S), y=ETF log return.

Joint Wald H0 = all PAST cause-lags 0; member FE (linear); hac-panel SE by member $\times$ day. p<0.05 $\Rightarrow$ leads.

1. CALENDAR-TIME

Direction A — Pooled $\rightarrow$ ETF:

$$y_t = \alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j} + \sum_{d=1..30} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags p=1 (BIC) ; cause block K=4 ; day-FE over 31 days ; hac-panel SE (bw=4).

H0: $b_1 = \dots = b_4 = 0 \rightarrow$ Wald $\chi^2(4) = 12.33$, p = 0.0151 ** | BH-FDR p = 0.0542 *

Direction B — ETF $\rightarrow$ Pooled:

$$x_t = \alpha + (\text{no } x \text{ self-lag term: BIC chose } p=0) + \sum_{j=1..4} d_j \cdot y_{t-j} + \sum_{d=1..30} \gamma_d \cdot \text{Day}_d$$

controls: ADL self-lags p=0 (BIC) ; cause block K=4 ; day-FE over 31 days ; hac-panel SE (bw=4).

H0: $d_1 = \dots = d_4 = 0 \rightarrow$ Wald $\chi^2(4) = 2.71$, p = 0.607 | BH-FDR p = 0.812

Direction A (probit, ETF up/down):

$$\Pr(y_t > 0) = \Phi(\alpha + \phi_1 \cdot y_{t-1} + \sum_{j=1..4} b_j \cdot x_{t-j})$$

controls: ADL self-lags p=1 (BIC) ; cause block K=4 ; sample spans 30 days (probit: no day-FE) ; hac-panel SE (bw=4).

H0: $b_1 = \dots = b_4 = 0 \rightarrow$ Wald $\chi^2(4) = 15.08$, p = 0.00453 *** | BH-FDR p = 0.0559 *

=> calendar verdict: Pooled Granger-leads ETF.

2. EVENT-TIME

(not estimable)